ALPHA COLLEGE OF ENGINEERING
THIRUMAZHISAI, CHENNAI
(Approved by AICTE and Affiliated to Anna University)

PROJECT REPORT

CUSTOMER SUPPORT TICKET PRIORITY PREDICTION AND AUTOMATED ASSIGNMENT SYSTEM USING AGENTFORCE

An AI-assisted Salesforce CRM solution for automated customer support ticket prioritization and assignment

Item	Details
Project Title	Customer Support Ticket Priority Prediction and Automated Assignment System Using Agentforce
Team ID	SWTID-2026-9909
Team Lead	DHIVYA SRI A (dhivyasri.abel@gmail.com)
Team Member	BHAVASHIKA R S (bhavashika04@gmail.com)
Team Member	BUVAHARINI M (bbuvaharini@gmail.com)
Team Member	PREETHI M (preethimurugan1403@gmail.com)
Team Member	SUGANYA A (rockysuganyasuganya@gmail.com)

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PROJECT SUMMARY

Customer Support Ticket Priority Prediction and Automated Assignment System Using Agentforce is a Salesforce-based solution designed to automate the analysis, prioritization, and assignment of customer support tickets. The system analyzes ticket descriptions and classifies them into High, Medium, or Low priority based on configured urgency conditions. The solution uses a Support Ticket Intelligence custom object to store customer support ticket information, including Account, Contact, Issue Type, Description, Priority Level, Status, Assigned Agent, and SLA Breach Risk. An Auto-Launched Flow retrieves the latest ticket associated with a customer Account and analyzes its description.

Agentforce is integrated with the Flow through a Support Ticket Priority Analysis subagent. When a user provides an Account Name, Agentforce retrieves the latest support ticket and determines its priority. High-priority tickets can automatically create an Urgent Ticket Handling task and assign the ticket to a senior support agent.

The system reduces manual ticket prioritization and assignment effort, helps support teams identify critical tickets quickly, and provides a conversational way to analyzes ticket information through Agentforce.



1. Introduction

1.1 Project Overview

The Customer Support Ticket Priority Prediction and Automated Assignment System Using Agentforce is a Salesforce-based solution that analyzes customer support tickets and classifies them as High, Medium, or Low priority. It uses the Support Ticket Intelligence object to manage ticket information.

An Auto-Launched Flow analyzes the latest ticket, creates an urgent task for High-priority tickets, and assigns the appropriate support level. Agentforce provides conversational ticket analysis and returns the priority and assignment details.

The main goal is to reduce manual effort, identify critical tickets faster, and improve support-team productivity.

1.2 Purpose

The system is designed to:

- Automatically classify support tickets based on priority.
- Retrieve and analyze the latest customer ticket.
- Create urgent tasks for High-priority tickets.
- Assign tickets to the appropriate support agent.
- Reduce manual prioritization and assignment.
- Improve response time and support-team productivity.

Expected Business Value

Business Outcome	Target
Reduced manual work	Automatic ticket analysis and assignment
Faster ticket handling	Quick identification of critical tickets
Improved productivity	Less repetitive support work
Decision making	Faster handling of urgent customer issues

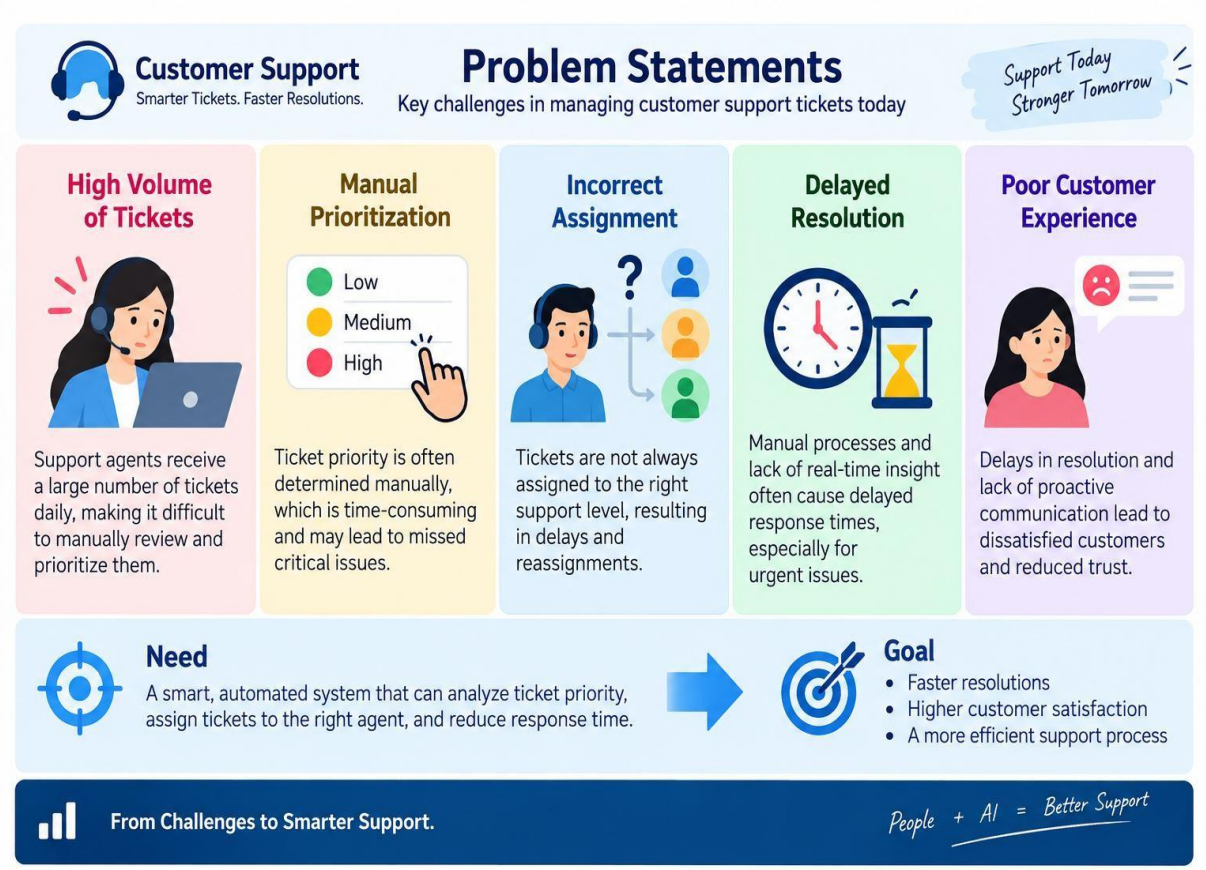
2. IDEATION PHASE

2.1 Problem Statements

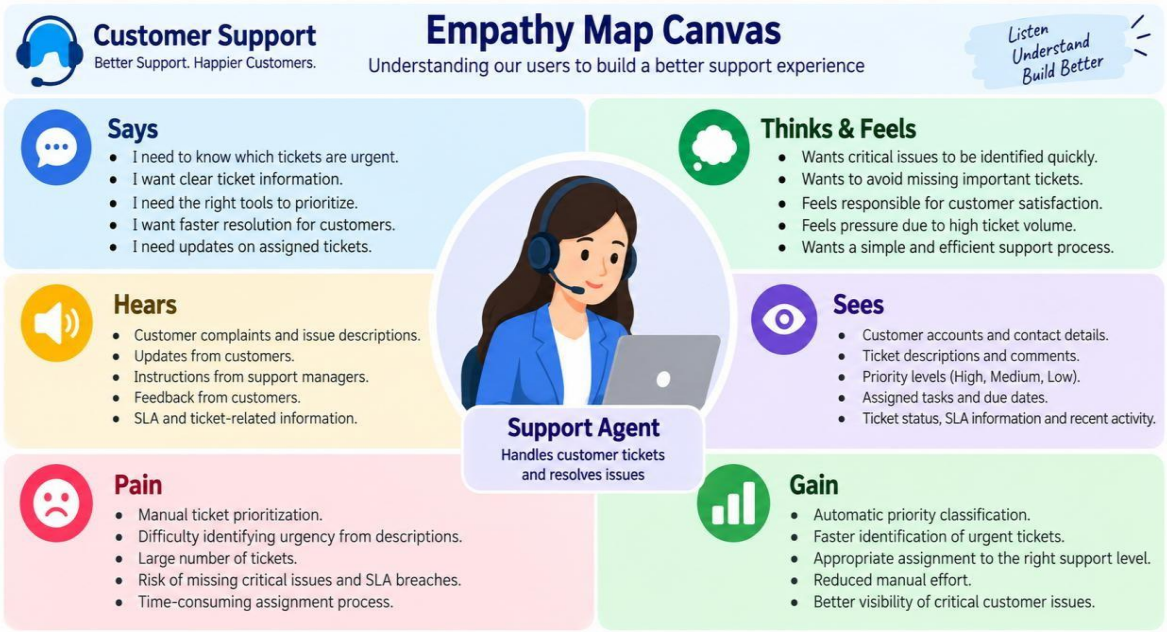
The following problem statements were identified from the viewpoints of users involved in customer support and ticket handling.

PS	Role	I am trying to	But	Because	Which makes me feel
PS-1	Support Agent	Identify urgent customer issues quickly	Important tickets may not be highlighted immediately	Tickets are received with different levels of urgency	Stressed
PS-2	Customer	Receive quick attention for serious problems	Critical requests may take longer to reach the right agent	Tickets are handled based on manual prioritization	Concerned
PS-3	Support Manager	Distribute tickets effectively	Choosing the appropriate support level can take time	Ticket information must be reviewed before assignment	Overloaded
PS-4	Support Agent	Understand the urgency of a ticket	Important terms in the description may be overlooked attention	Customer descriptions vary in detail and wording	Unsure
PS-5	Support Manager	Monitor unresolved support issues	Older tickets may require additional	SLA-related information needs continuous monitoring	Worried

Customer support teams may face difficulties when large numbers of tickets need to be reviewed, prioritized and assigned manually. A solution is therefore required to identify ticket urgency, support faster assignment and reduce repetitive manual activities.



2.2 Empathy Map Canvas

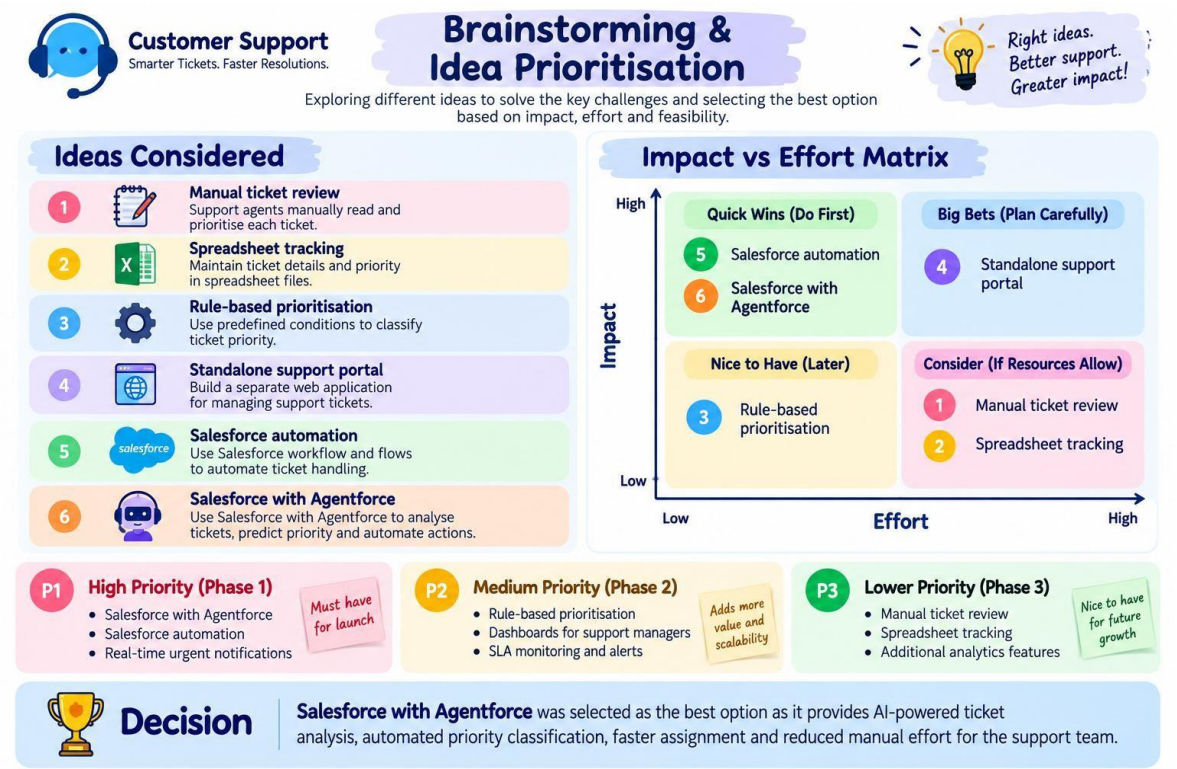


User Persona: Support Agent

Main responsibility: Reviewing customer tickets, identifying urgent issues, handling assigned cases and supporting timely resolution.

- Thinks & Feels: Wants important customer issues to be identified quickly and resolved without unnecessary delays.
- Hears: Customer complaints, ticket descriptions, updates from customers and instructions from support managers.
- Sees: Customer accounts, ticket descriptions, issue types, priority levels, assigned agents, task information and SLA details.
- Says & Does: Reviews ticket information, handles assigned cases, checks urgent requests and communicates with customers or team members.
- Pain: Manual priority checking, difficulty recognizing urgency from different descriptions, high ticket volume and possibility of delayed handling.
- Gain: Automatic priority prediction, faster identification of important tickets, appropriate assignment, automatic urgent-task creation and reduced manual effort.

2.3 Brainstorming & Idea Prioritization



Options Considered

Option	How it would work	Main Limitation
Manual Ticket Review	Support agents examine each ticket and decide its priority	Time-consuming for large ticket volumes
Spreadsheet Tracking	Ticket information and priority are maintained in spreadsheets	Limited automation and real-time processing
Rule-Based System	Predefined conditions are used to classify ticket priority	Less flexible for conversational interaction
Standalone Support Portal	A separate application manages support tickets	Requires additional development and maintenance
Salesforce with Agentforce	Salesforce Flow and Agentforce analyze tickets and automate support actions	Requires proper Salesforce configuration

Why Salesforce with Agentforce

- Automation – Flow can automate ticket analysis, priority classification and task creation.
- AI-assisted interaction – Agentforce provides conversational access to ticket analysis.
- Faster prioritization – Tickets can be categorized based on configured urgency conditions.
- Reduced manual work – High-priority tickets can trigger an urgent handling task.
- Centralized data – Customer and ticket information can be maintained within Salesforce.

Decision :

Salesforce with Agentforce was selected as the proposed platform for implementing the Customer Support Ticket Priority Prediction and Automated Assignment System.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey

Stage	Customer Journey Step	What Happens
1	Customer & Account Identification	Customer account and related support information are identified in Salesforce.
2	Ticket Creation	A support ticket is created with issue type and description.
3	Ticket Analysis	The ticket description and available information are reviewed for urgency.
4	Priority Classification	The system determines whether the ticket is High, Medium or Low priority.
5	Agent Assignment	The ticket is assigned to a suitable support agent based on its priority.
6	Urgent Ticket Handling	High-priority tickets can generate an urgent handling task for senior support staff.
7	SLA Monitoring	SLA breach risk is tracked to identify tickets requiring attention.
8	Resolution & Closure	The support team handles the issue and updates the ticket status.



3.2 Solution Requirements

ID	Requirement
FR-1	Customer account and contact management
FR-2	Support ticket creation and storage
FR-3	Ticket description and issue-type management
FR-4	Automatic ticket priority classification
FR-5	High, Medium and Low priority identification
FR-6	Automated support-agent assignment
FR-7	Urgent task creation for critical tickets
FR-8	SLA breach-risk identification
FR-9	Ticket status management
FR-10	Agentforce-based ticket analysis
FR-11	Automated support workflow using Flow
FR-12	Ticket information retrieval through Agentforce

Non-Functional Requirements

Attribute	Description
Usability	Simple interface for support staff to view and manage tickets.
Security	Access to ticket information is controlled through Salesforce permissions.
Reliability	Automation should consistently classify and process tickets.
Performance	Ticket analysis and assignment should complete efficiently.
Availability	Salesforce provides cloud-based access to authorized users.
Scalability	The system can support increasing numbers of customers and tickets.
Maintainability	Flows and Salesforce configurations can be updated when requirements change.
Data Integrity	Ticket and customer information should remain accurate and consistent.

3.3 Data Flow

Main Flow -

Customer Account → Support Ticket Creation → Ticket Description → Support Ticket Intelligence → Automated Flow → Ticket Priority Analysis → Priority Classification → Agent Assignment → Urgent Task / Normal Handling → Status Update.

Supporting Flows -

High-Priority Flow:

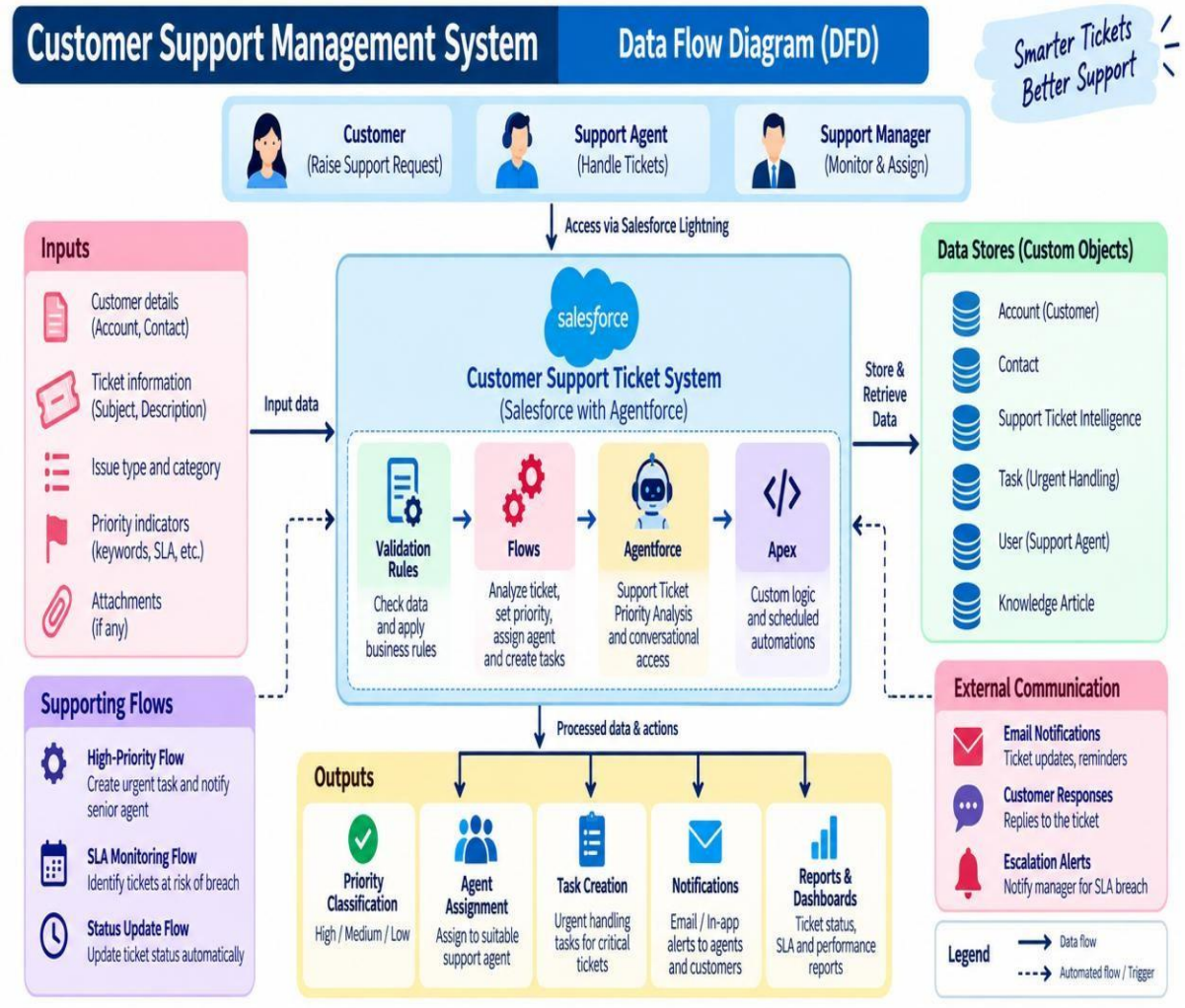
Ticket received → Priority identified as High → Urgent Ticket Handling Task created → Senior Support Agent assigned → Ticket handled quickly.

Agentforce Flow:

User provides Account Name → Agentforce retrieves the latest support ticket → Support Ticket Priority Analysis subagent analyses the information → Priority and assigned agent are returned → Action message is displayed.

Overview -

Users → Salesforce → Support Ticket Intelligence → Flow & Agentforce → Priority Classification → Assignment → Task Creation → Ticket Resolution.



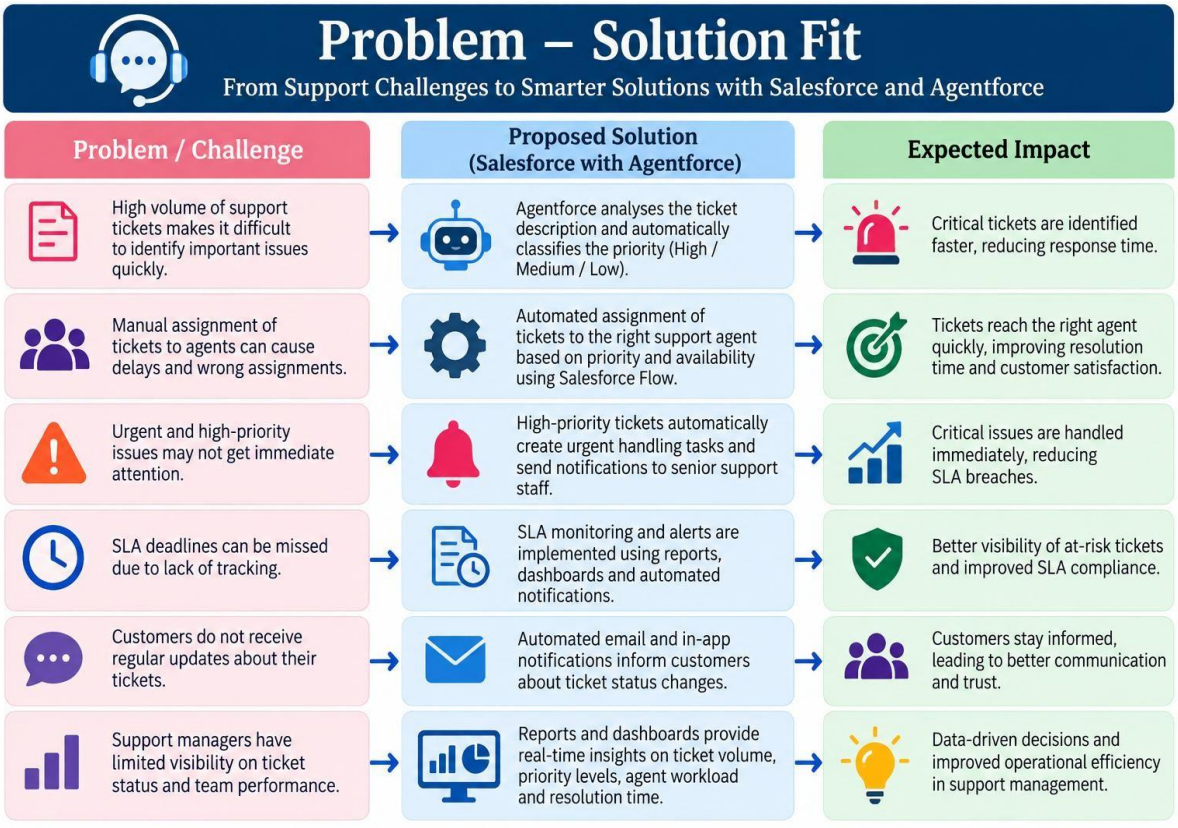
3.4 Technology Stack

Layer	Technology
Platform	Salesforce CRM
User Interface	Salesforce Lightning Experience
Data Storage	Custom Salesforce Object – Support Ticket Intelligence
Automation	Salesforce Flow
AI / Conversational Layer	Agentforce
Agentforce Component	Support Ticket Priority Analysis
Business Logic	Salesforce configuration and Flow logic
Priority Levels	High, Medium, Low
Security	Salesforce Profiles, Roles and Permissions
Reporting	Salesforce Reports and Dashboards

4. PROJECT DESIGN

4.1 Problem–Solution Fit

Problem	Proposed Solution
Support agents manually review many tickets	Automated ticket analysis using Flow and Agentforce
Important tickets may not be identified quickly	Automatic priority classification
Manual assignment can delay ticket handling	Automated assignment based on ticket priority
Urgent issues may require immediate action	Automatic urgent handling task creation
Ticket urgency can be missed in descriptions	Agentforce-assisted ticket analysis
SLA risks may not be noticed early	SLA breach-risk information maintained with ticket
Support managers need better visibility	Centralized Salesforce ticket information



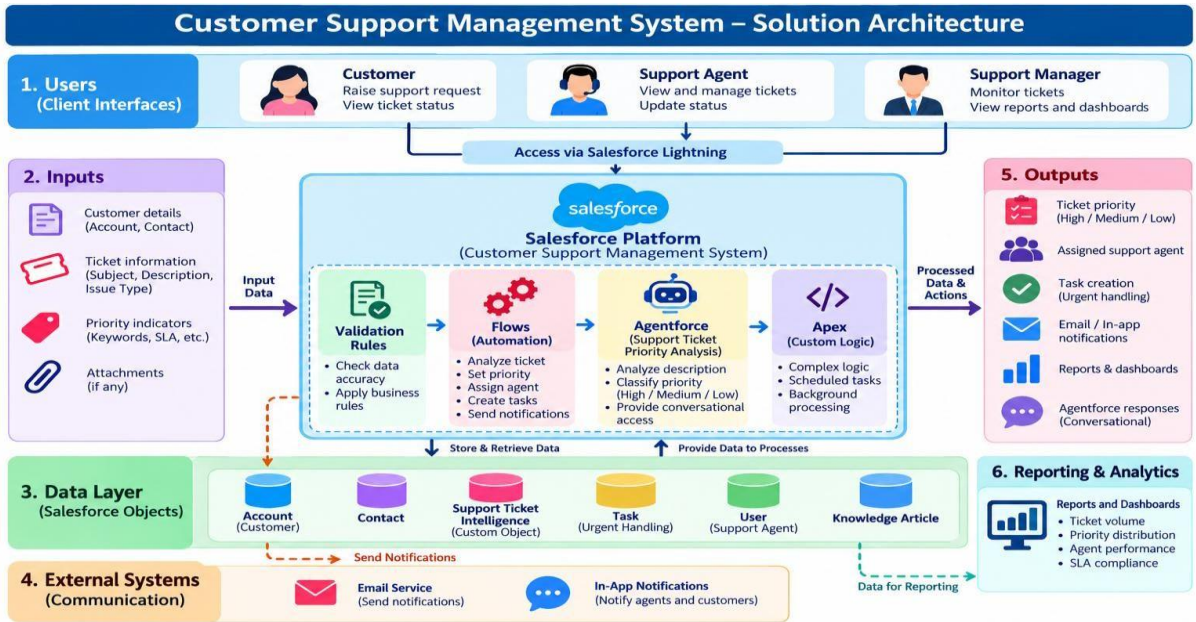
4.2 Proposed Solution

Aspect	Description
Problem	Manual ticket prioritization and assignment can delay customer support.
Solution	Salesforce-based ticket intelligence and automated assignment system.
Innovation	Agentforce-assisted analysis of customer support tickets.
Automation	Flow automatically processes ticket information and performs required actions.
Priority	Tickets are classified as High, Medium or Low.
Assignment	Suitable support agents can be assigned according to ticket priority.
Impact	Faster identification of critical issues and reduced manual effort.
Scalability	The system can handle increasing customer support ticket volumes.

4.3 Solution Architecture

Components

Users / Support Team → Agentforce → Support Ticket Priority Analysis → Auto-Launched Flow → Account & Ticket Data → Description Analysis → Priority Decision (High / Medium / Low) → Task / Agent Assignment → Final Ticket Information → Agentforce Response



5. Project Planning & Scheduling

5.1 Project Planning

- **Agile methodology**
- Work organised as **Epics → User Stories → Story Points**
- **Sprint-based development**
- Sprint progress tracked through completed story points
- Team capacity considered while planning sprint activities

Product Backlog :

Sprint	Epic	Story	Description	Pts	Priority	Owner
1	Ticket Management	US-01	As a Support Agent, I want to create and maintain support ticket details so customer issues can be tracked systematically.	3	High	DHIVYA SRI.A
1	Data Management	US-02	As a Support Agent, I want customer Account and Contact information linked to tickets so ticket details are available in one place.	5	High	BHAVASHIKA.R.S
2	Ticket Analysis	US-03	As a Support Agent, I want ticket descriptions to be analyzes so the urgency of customer issues can be identified.	5	High	BUVAHARINI.M
2	Priority Classification	US-04	As a Support Manager, I want tickets classified as High, Medium or Low so critical issues can be handled first.	5	High	PREETHI.M
3	Assignment & Automation	US-05	As a Support Manager, I want High-priority tickets assigned to the appropriate support level so urgent	5	High	SUGANYA.A

			issues receive faster attention.			
3	Task Automation	US-06	As a Support Agent, I want an urgent handling task created for High-priority tickets so critical cases are not missed.	3	High	DHIVYA SRI.A
4	Agentforce	US-07	As a Support User, I want to provide an Account Name to Agentforce and receive ticket priority and assignment information.	5	High	BHAVASHIKA.R.S
4	SLA & Monitoring	US-08	As a Support Manager, I want SLA breach risk and ticket information to be available so unresolved issues can be monitored.	5	Medium	BUVAHARINI.M

Sprint Tracker :

Sprint	Points	Duration	Start	Planned End	Completed	Released
Sprint-1	20	6 days	28/9/26	03/09/26	20	03/09/26
Sprint-2	20	6 days	5/10/26	10/10/26		
Sprint-3	20	6 days	12/10/26	17/09/26		
Sprint-4	20	6 days	19/10/26	24/10/26		

Velocity:
The team planned 20 story points per sprint, giving a planned capacity of 80 story points across four 6-day sprints. The average velocity is 20 story points per sprint, or approximately 3.33 story points per day (20 ÷ 6)

6. PROJECT DEVELOPMENT PHASE

The development phase focuses on implementing the Customer Support Ticket Priority Prediction and Automated Assignment System using Salesforce and Agentforce. The implementation includes Salesforce environment setup, custom object configuration, Auto-Launched Flow development, Agentforce subagent configuration, and final system integration.

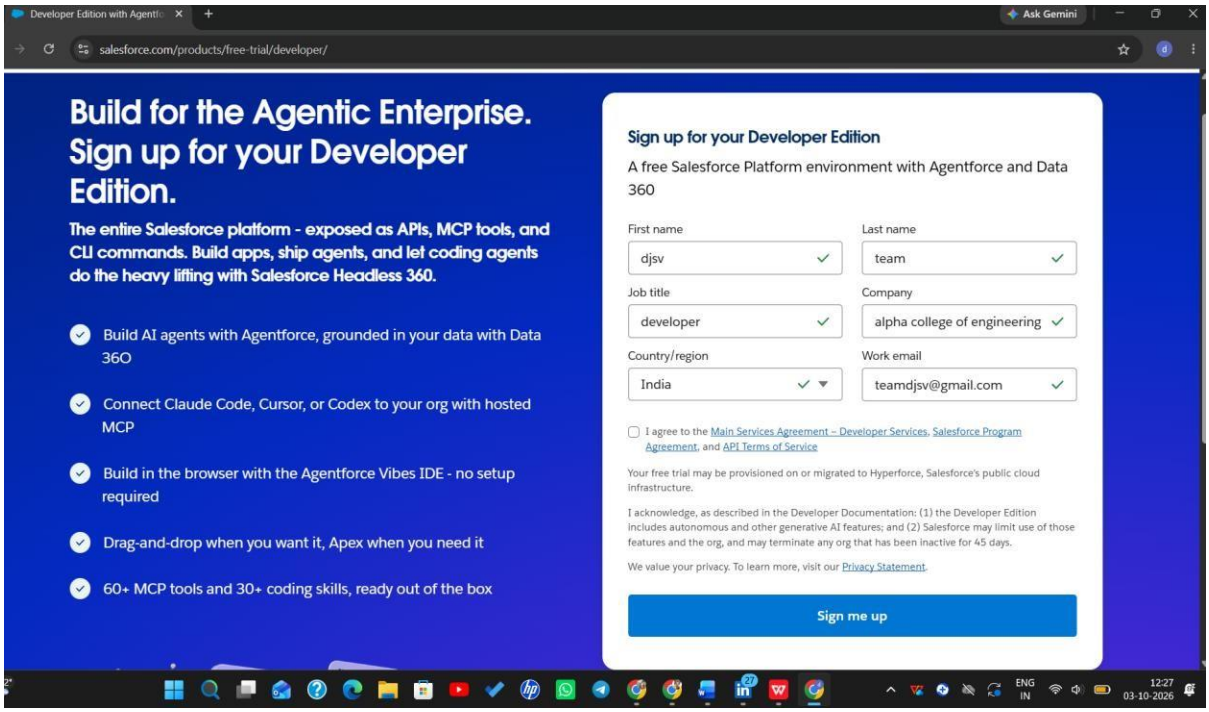
6.1 Phase 1: Salesforce Environment Setup & Project Configuration

Step 1: Salesforce Developer Account

A Salesforce Developer Environment was created and accessed to provide the required platform for developing the Customer Support Ticket Priority Prediction and Automated Assignment System. The environment was used for configuring custom objects, fields, automation flows, Agentforce components, and testing the complete solution.

Activities:

1. Created and accessed the Salesforce Developer Account.
2. Logged into the Salesforce Developer Environment.
3. Verified the Salesforce Setup and development workspace.
4. Prepared the environment for custom object, Flow, and Agentforce configuration.

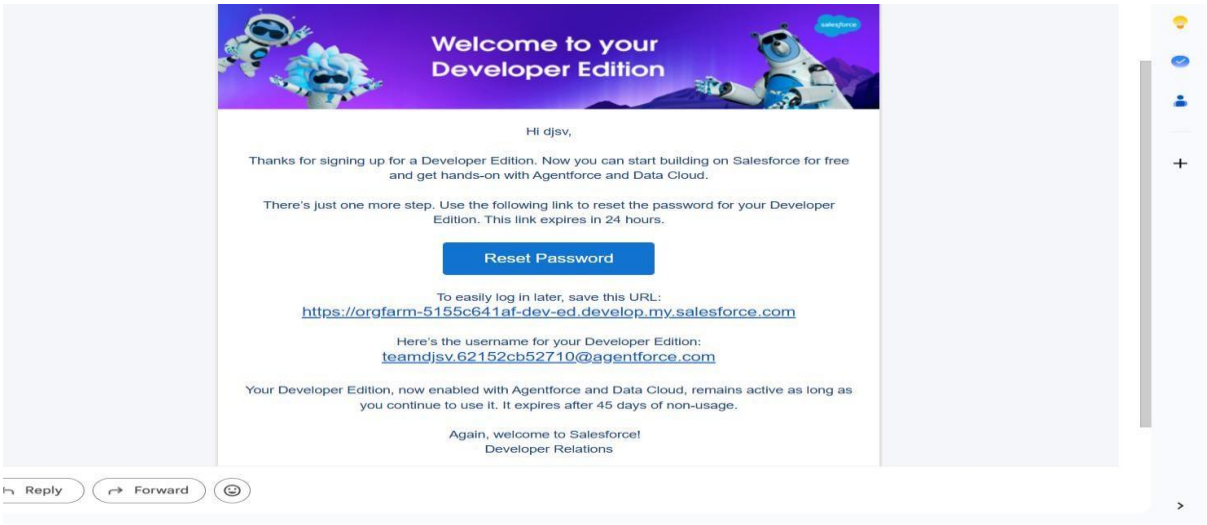


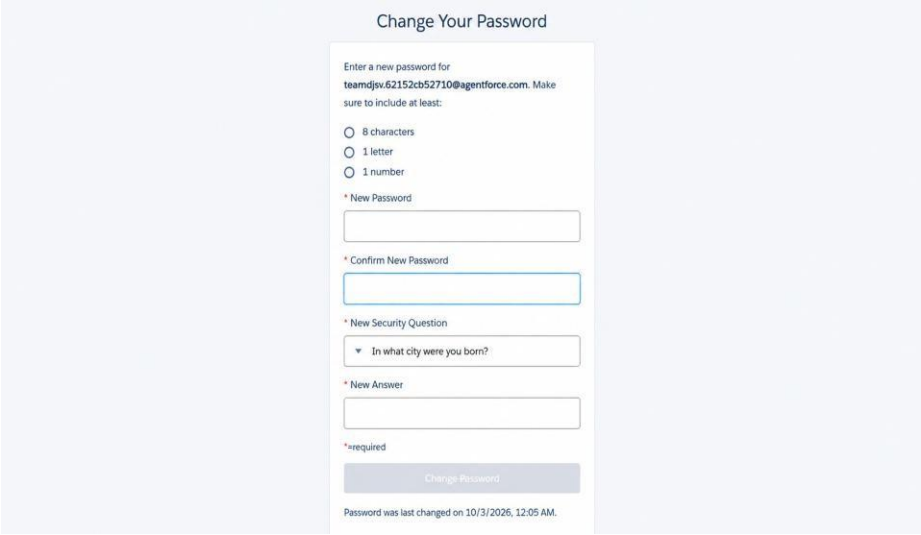
Step 2: Salesforce Environment Setup

The Salesforce Developer Environment was configured as the primary development workspace for the project. The required Salesforce setup components were accessed and prepared for creating custom objects, fields, automation flows, and Agentforce components.

Activities:

1. Accessed the Salesforce Setup environment.
2. Verified the required Salesforce configuration options.
3. Prepared the environment for creating the project-specific custom object and fields.
4. Configured the development workspace required for Flow and Agentforce implementation.



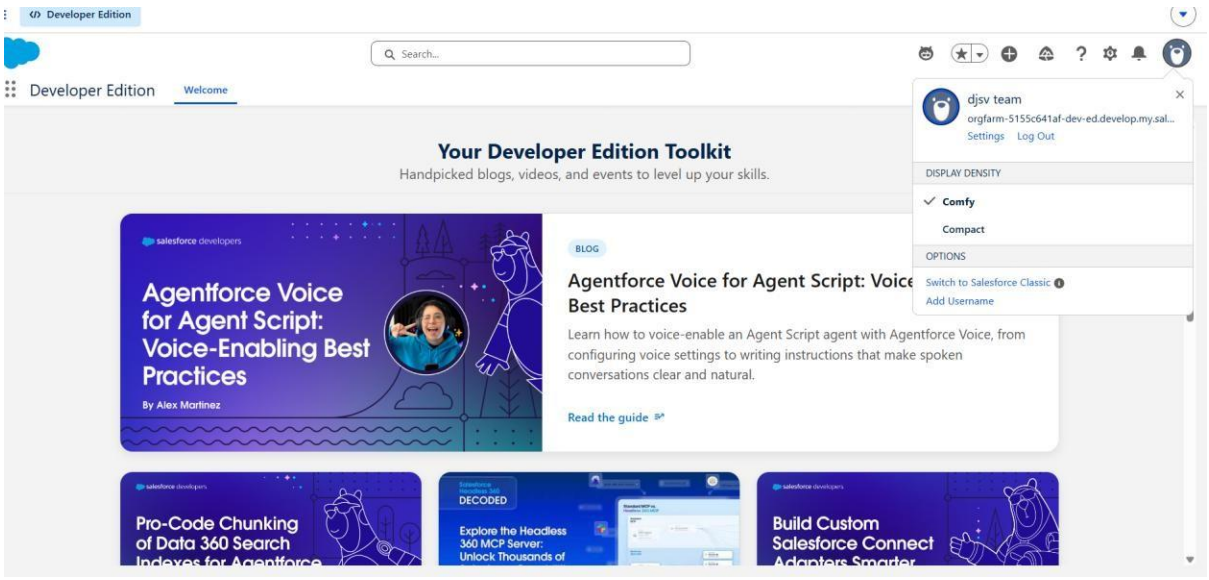


Step 3: Agentforce Enablement

Agentforce was enabled in the Salesforce Developer Environment to support the development of the intelligent customer support ticket automation system. The Agentforce configuration provides the required platform for creating and configuring the **Support Ticket Priority Analysis** subagent and connecting it with the project automation flow.

Activities:

1. Accessed Agentforce configuration from Salesforce Setup.
2. Enabled the required Agentforce features.
3. Verified that Agentforce was available in the Salesforce environment.
4. Prepared the environment for creating the Support Ticket Priority Analysis subagent.



6.2 Phase 2: Data Modeling & Custom Object Configuration

Step 1: Create/Configure Custom Object

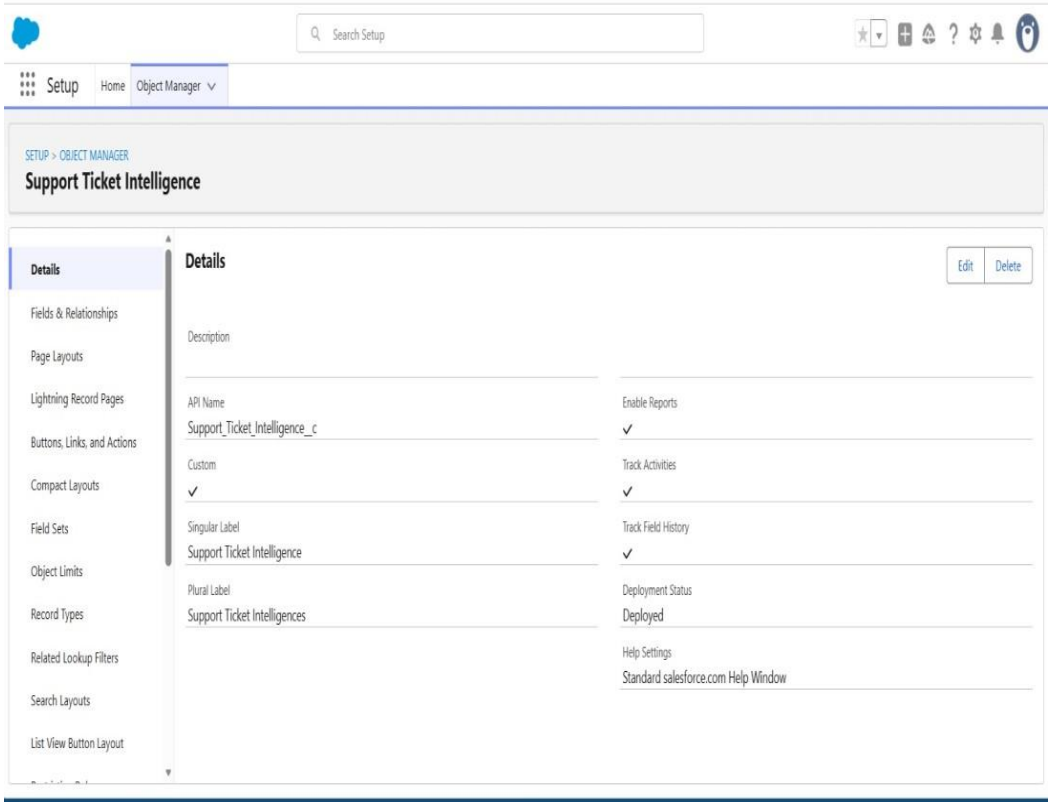
Support Ticket Intelligence Custom Object

A custom object named Support Ticket Intelligence was created in Salesforce to store and manage customer support ticket information required for priority prediction and automated assignment. The custom object serves as the central data structure for connecting ticket information with the automation Flow and Agentforce.

Activities:

1. Created the Support Ticket Intelligence custom object.

2. Configured the object label and API name.
3. Enabled the required object settings.
4. Prepared the custom object for adding project-specific fields.
5. Verified the custom object in Salesforce Object Manager.

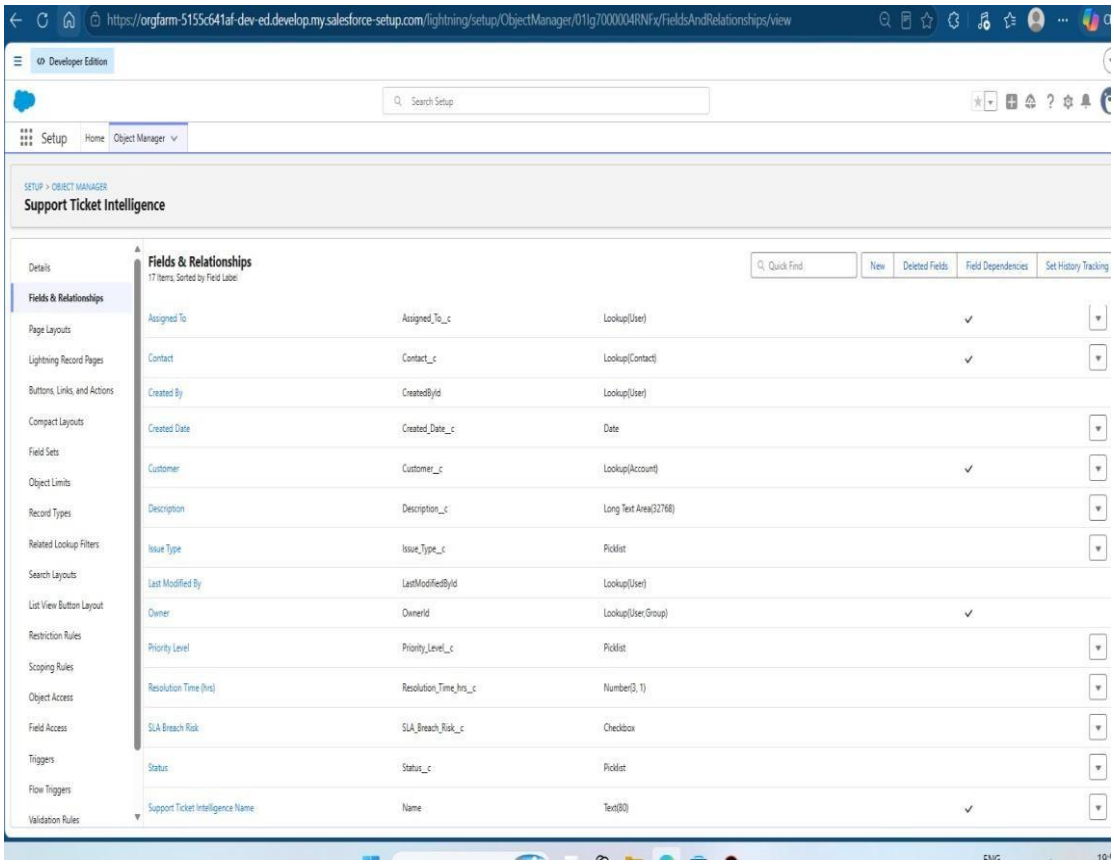


Step 2: Fields and Relationships

The required fields and relationships were configured in the **Support Ticket Intelligence** custom object to store the information needed for ticket analysis, priority classification, and automated assignment. These fields provide the input data required by the Flow and Agentforce components.

Activities:

1. Opened the **Support Ticket Intelligence** custom object.
2. Created and configured the required project-specific fields.
3. Configured appropriate field data types.
4. Added the required relationships with Salesforce records.
5. Verified the configured fields in the object.

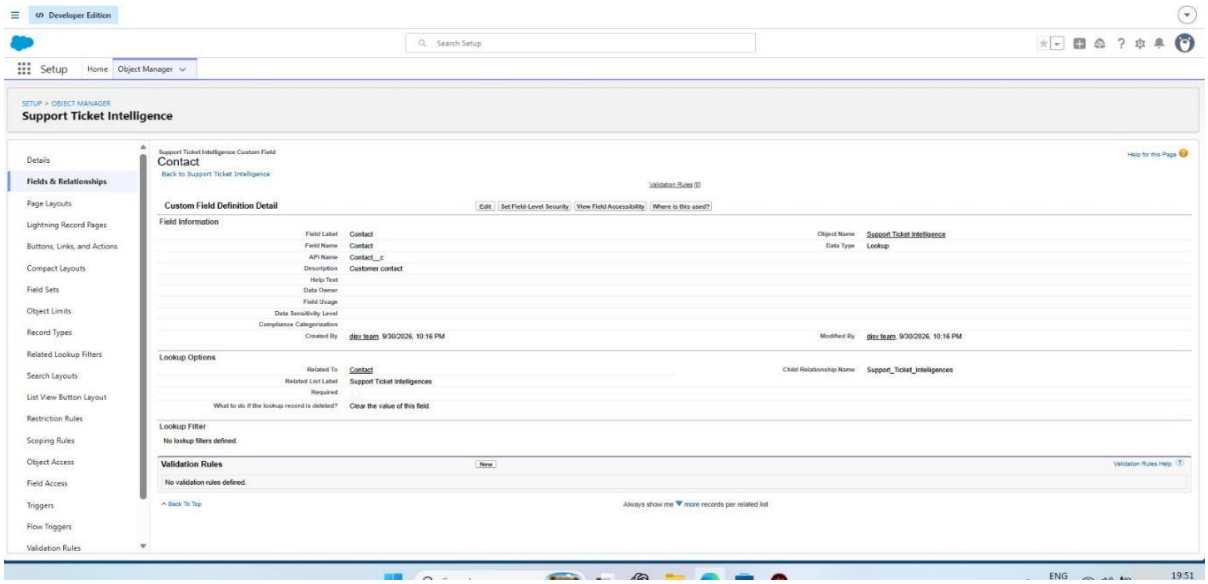


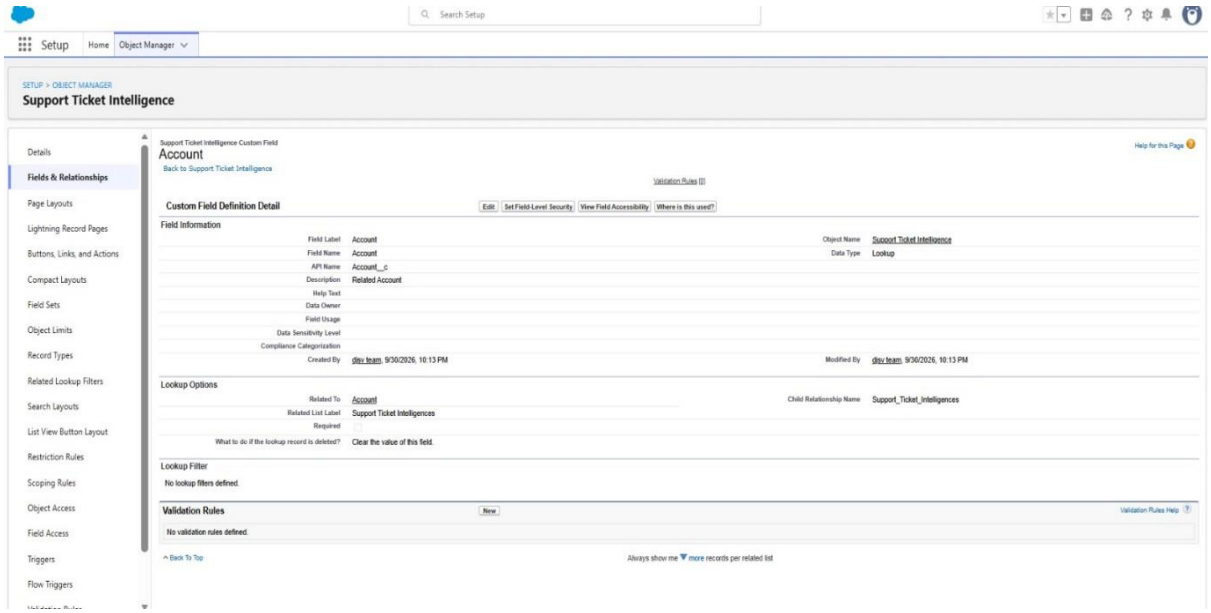
Step 3: Account and Contact Association

The **Support Ticket Intelligence** object was associated with the required **Account and Contact** records to maintain customer-related information for each support ticket. This association enables the system to identify the customer account and contact details during ticket processing and automation.

Activities:

1. Configured the **Account** relationship in the Support Ticket Intelligence object.
2. Configured the required **Contact** relationship.
3. Associated support ticket records with the corresponding customer Account and Contact.
4. Verified the relationships through the created Salesforce records.





Step 4: User and Task Records

The required **User** and **Task** records were configured in Salesforce to support automated ticket assignment and follow-up activities. User records represent the support agents, while Task records are used to track actions generated for high-priority support tickets.

Activities:

1. Verified the available Salesforce User records.
2. Identified the appropriate support user for ticket assignment.

3. Configured Task records for follow-up activities.
4. Verified the relationship between the assigned user and generated tasks.
5. Prepared the records for integration with the automation Flow.

6.3 Phase 3 – Auto-Launched Flow Development & Automation

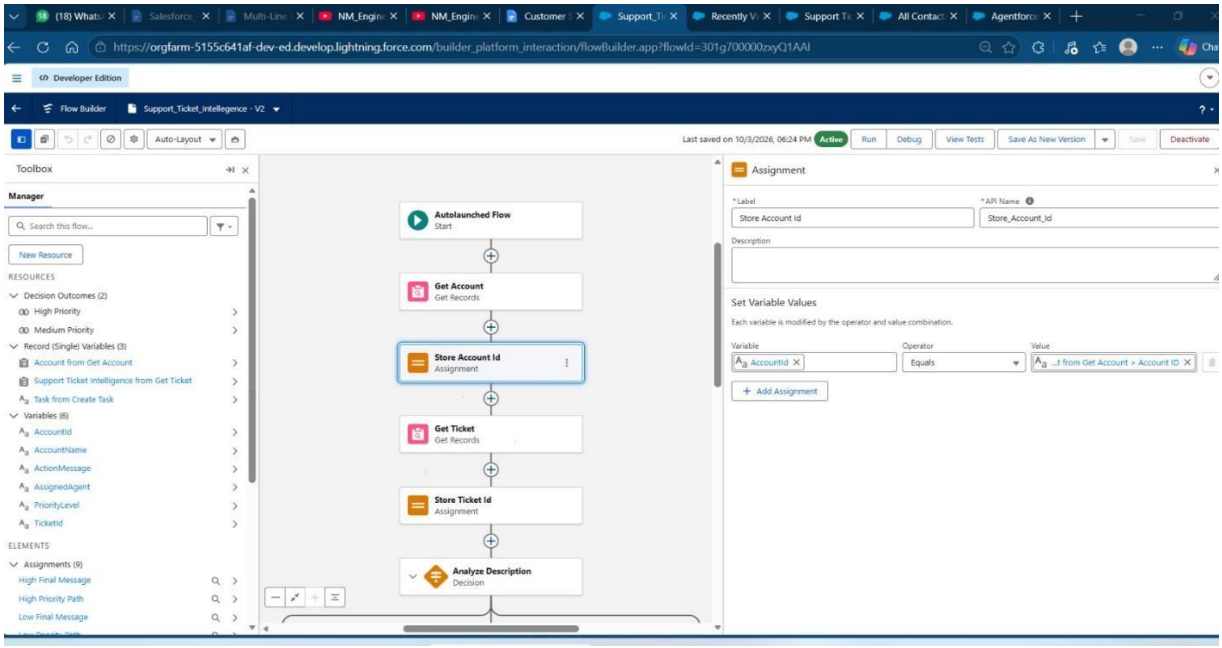
Step 1: Flow Creation & Flow Variables

Flow Variables

An Auto-Launched Flow was created to automate the processing of support ticket information. Input variables were configured to receive the required ticket and customer information during Flow execution. These variables allow the Flow to retrieve records, analyses ticket details, classify priority, create tasks, and assign the ticket to the appropriate support user.

Activities:

1. Created a new Auto-Launched Flow.
2. Configured the required input variables.
3. Defined the data types and availability of the variables.
4. Prepared the variables for use in subsequent Flow elements.
5. Saved the Flow for further development.

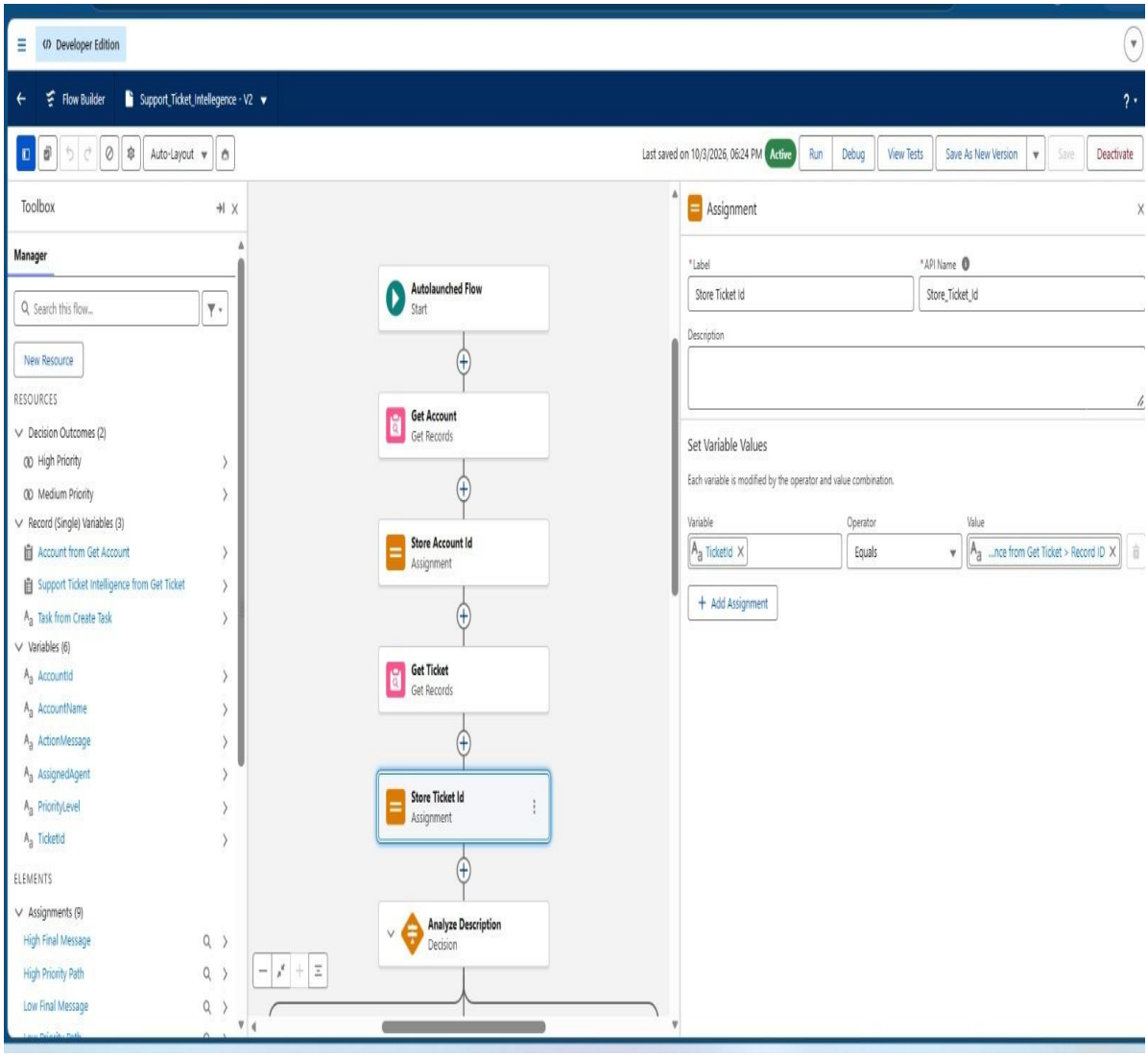


Step 2: Account Retrieval

The Flow was configured to retrieve the relevant Account record associated with the support ticket. This step ensures that the customer account information is available for subsequent ticket analysis and processing.

Activities:

1. Added a Get Records element to the Flow.
2. Selected the Account object for record retrieval.
3. Configured the required conditions using the Account information associated with the support ticket.
4. Stored the retrieved Account record for use in subsequent Flow elements.
5. Verified the Account retrieval configuration.

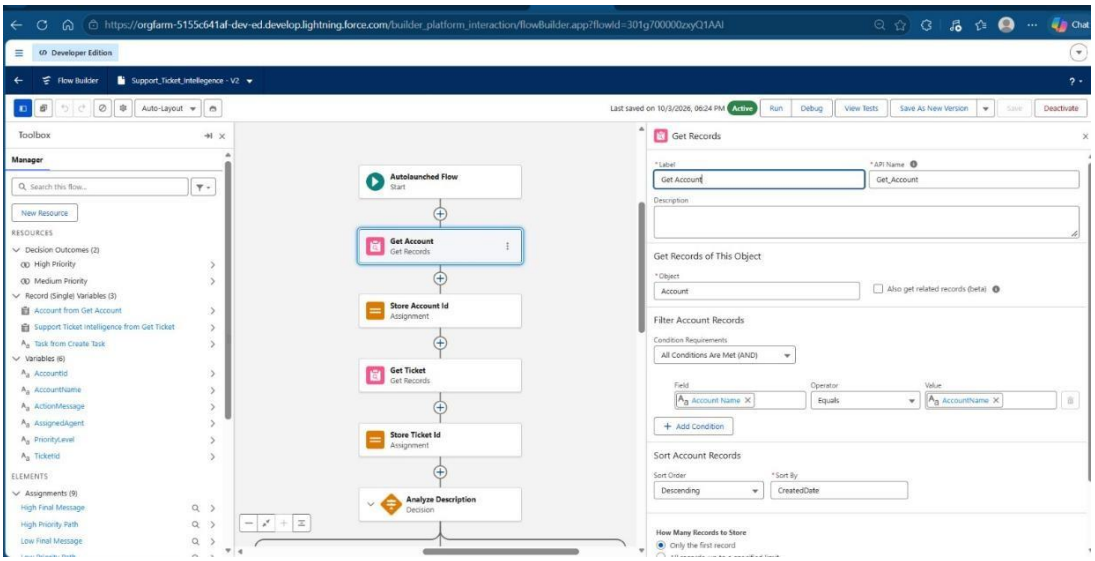


Step 3: Latest Ticket Retrieval

The Flow was configured to retrieve the latest relevant support ticket record from the Support Ticket Intelligence custom object. This step ensures that the most recent ticket information is available for description analysis and priority classification.

Activities:

1. Added a Get Records element to the Flow.
2. Selected the Support Ticket Intelligence object.
3. Configured the required filter conditions to identify the relevant ticket.
4. Configured the Flow to retrieve the latest matching ticket record.
5. Stored the retrieved ticket information for further processing.



Step 4: Ticket Description Analysis

The retrieved support ticket description was processed to identify the nature and severity of the customer's issue. The ticket description serves as the primary input for determining the appropriate priority level.

Activities:

1. Retrieved the ticket description from the Support Ticket Intelligence record.
2. Passed the description as input for analysis.
3. Evaluated the issue details and severity indicators in the description.
4. Prepared the analyzed information for priority classification.

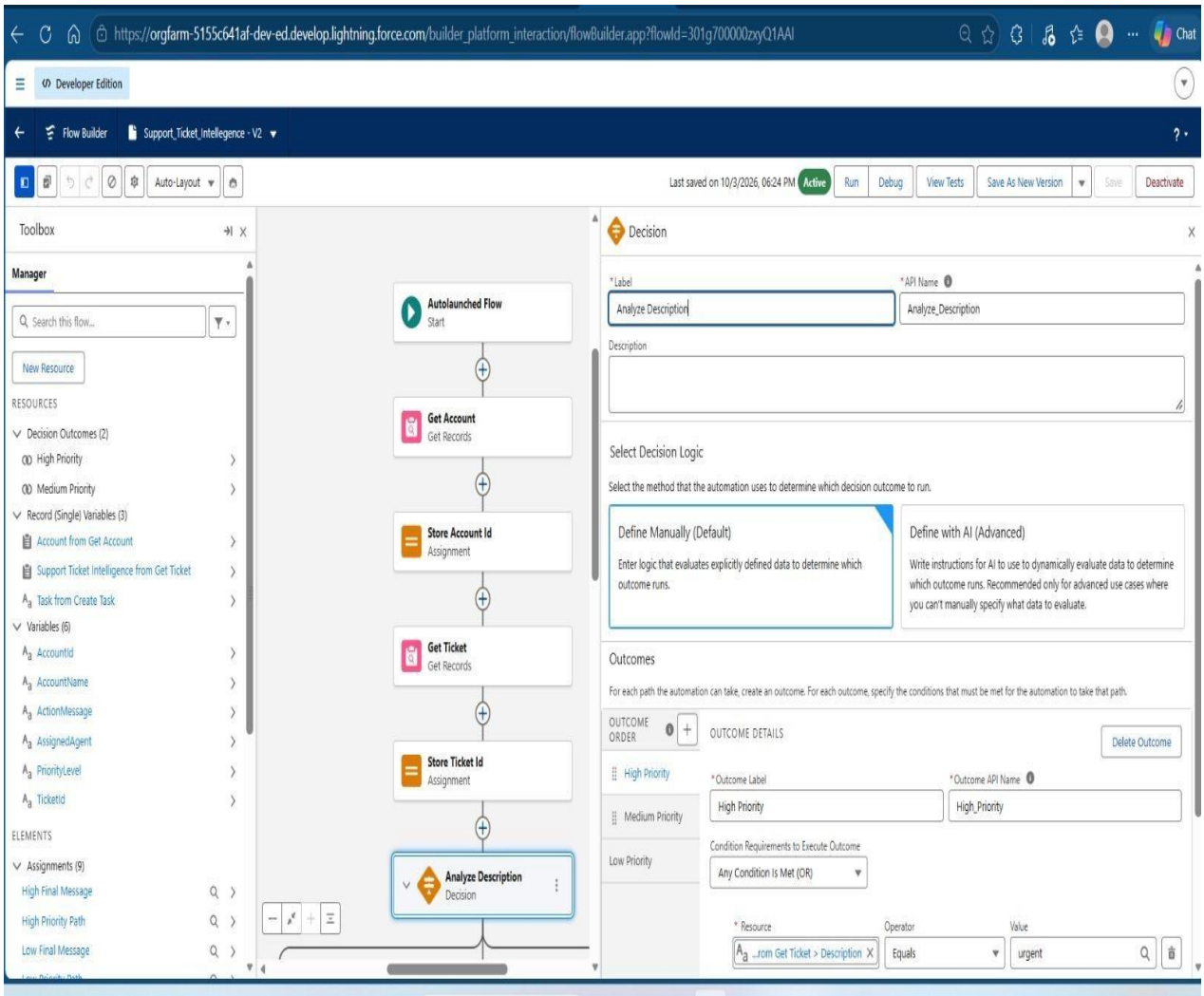
5. Verified that the ticket description was available for the next Flow step

Step 5: Priority Classification

The Flow was configured to classify the support ticket according to its priority based on the analyzed ticket information. The classification helps the system determine the appropriate response and follow-up action for the support request.

Activities:

1. Processed the analyzed ticket information.
2. Applied the configured priority classification logic.
3. Classified the support ticket into the appropriate priority level.
4. Stored the resulting priority value for subsequent automation steps.
5. Verified the priority classification output.



Step 6: High-Priority Task Creation

The Flow was configured to automatically create a Task when a support ticket is identified as high priority. This enables the support team to receive a follow-up task for critical customer issues and helps ensure that important tickets are handled promptly.

Activities:

1. Evaluated the ticket priority obtained from the classification step.
2. Identified tickets classified as High Priority.
3. Added a Create Records element to the Flow.
4. Configured the Task record with the required ticket and customer information.
5. Created the Task automatically for high-priority support tickets.
6. Verified the generated Task record in Salesforce.

Step 7: Agent Assignment

The Flow was configured to assign the support ticket and related task to the appropriate Salesforce support user. This automation helps ensure that the identified ticket is routed to the designated agent for further action.

Activities:

1. Identified the Salesforce User required for ticket assignment.
2. Retrieved or referenced the appropriate User record.
3. Assigned the support ticket to the selected support agent.
4. Associated the generated Task with the assigned User.
5. Verified the agent assignment in the Salesforce record.

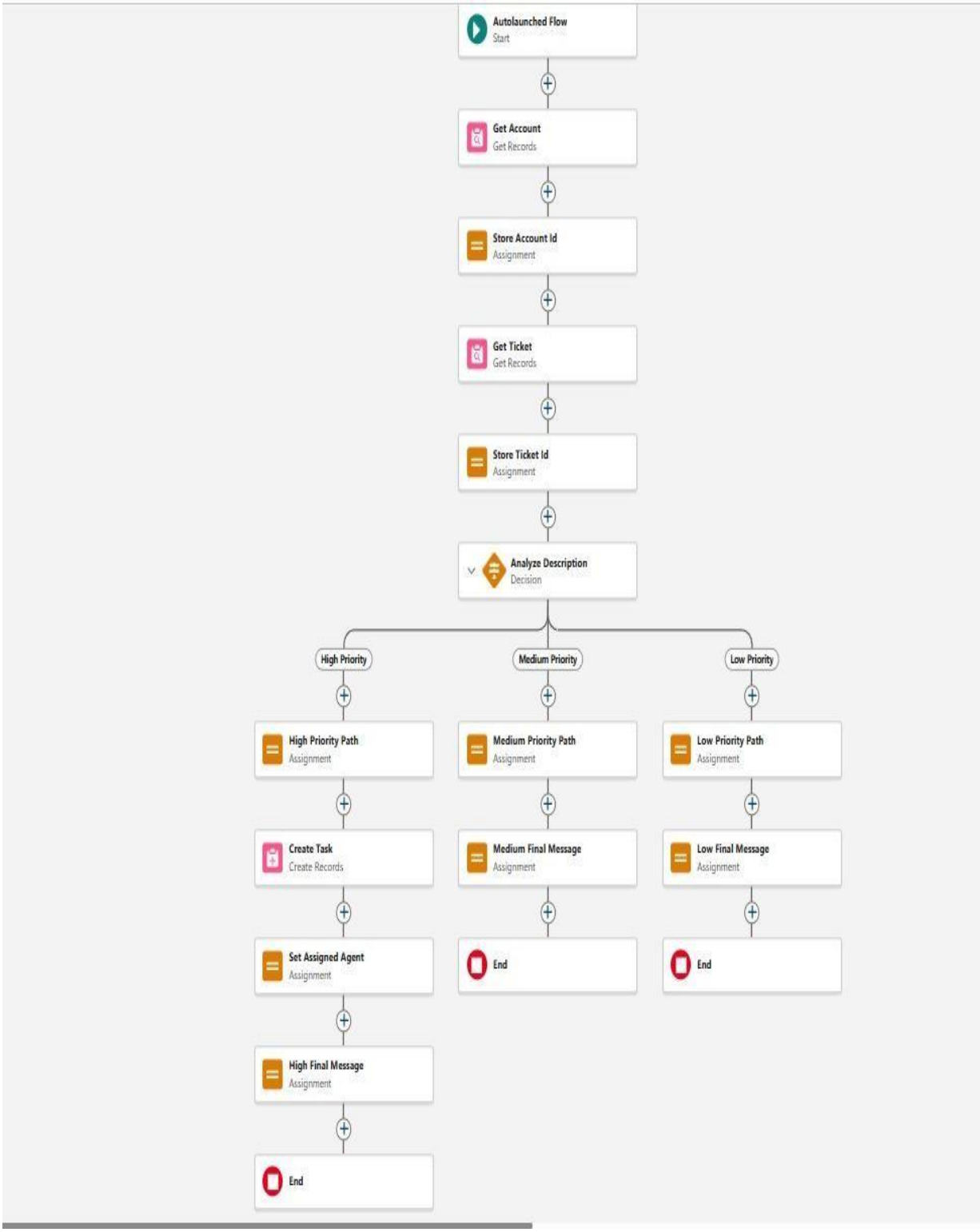
Step 8: Final Response Message

The final response message was configured to provide the result of the ticket processing after the Flow completes its automation steps. The response communicates the identified ticket priority and confirms the corresponding automated action.

Activities:

1. Prepared the final output message after ticket processing.
2. Included the relevant ticket priority information.
3. Confirmed the automated action performed by the Flow.
4. Configured the final response to be returned after Flow execution.

5. Verified the final output message.



6.4 Phase 4 – Step 1: Support Ticket Priority Analysis Subagent

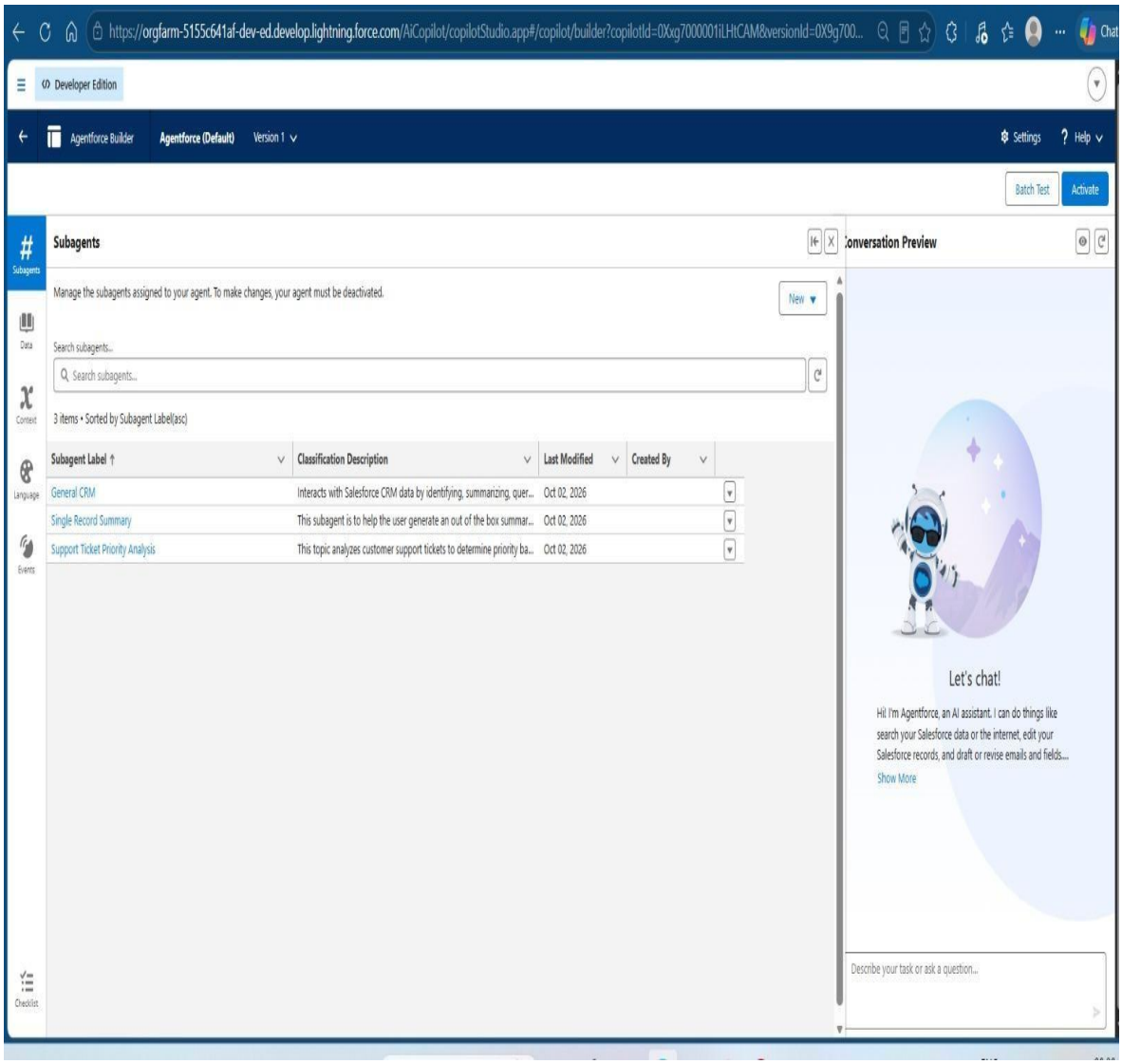
Step 1: Support Ticket Priority Analysis Subagent

Support Ticket Priority Analysis Subagent

A dedicated Agentforce subagent named Support Ticket Priority Analysis was configured to support the analysis of customer support tickets. The subagent is designed to process ticket information and determine the appropriate priority based on the configured instructions and automation logic.

Activities:

1. Accessed the Agentforce configuration environment.
2. Created the Support Ticket Priority Analysis subagent.
3. Configured the subagent name and required details.
4. Prepared the subagent for integration with the project Flow.
5. Verified that the subagent was available for further configuration.



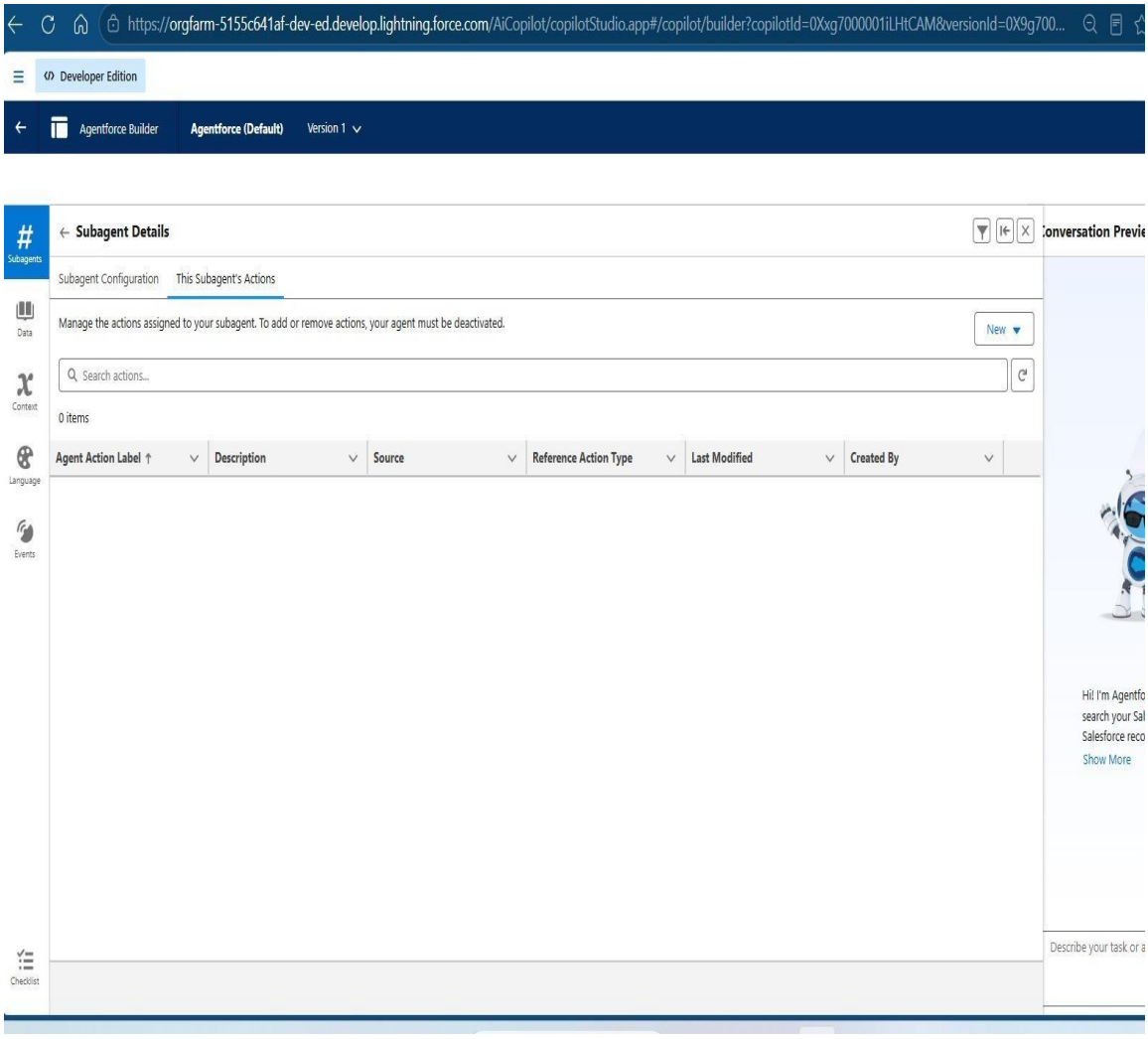
Step 2: Subagent Instructions

Subagent Instructions

Instructions were configured for the Support Ticket Priority Analysis subagent to guide how customer support ticket information should be processed and how the appropriate priority should be identified. The instructions ensure consistent processing of ticket information before the result is returned to the Salesforce automation.

Activities:

1. Defined the purpose of the Support Ticket Priority Analysis subagent.
2. Configured instructions for analyzing support ticket information.
3. Specified the priority classification requirements.
4. Defined the expected response format.
5. Verified the configured instructions before connecting the subagent with the Flow

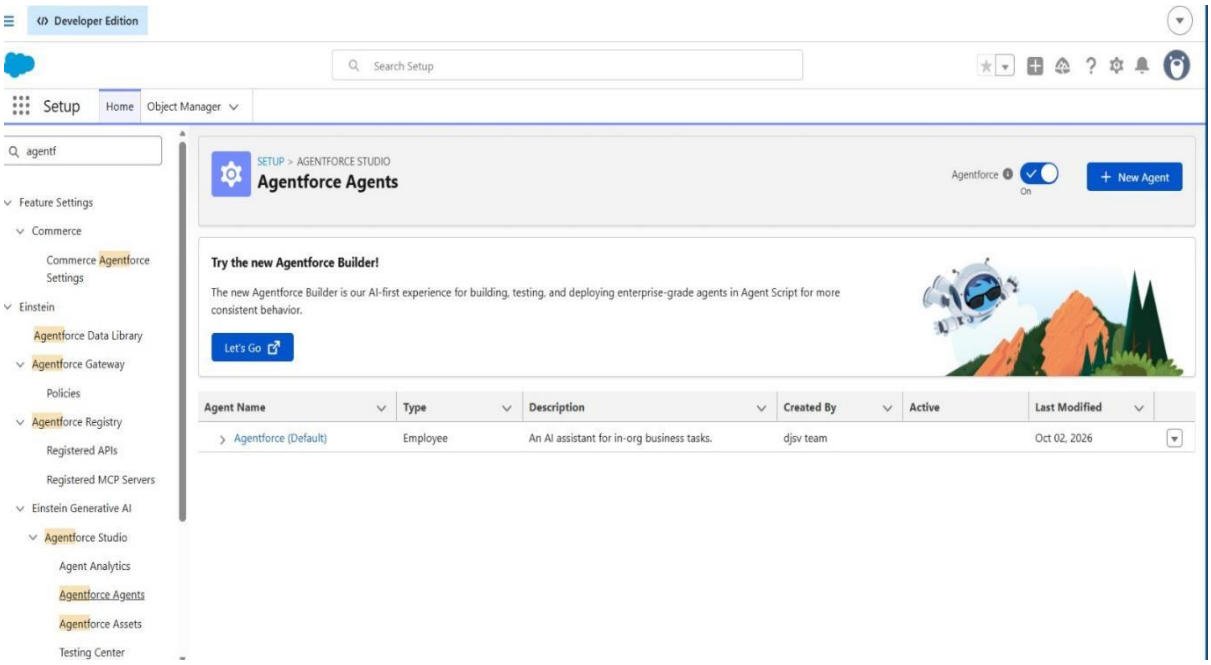


Step 3: Flow-Based Agentforce Action

A Flow-based Agentforce action was configured to connect the Support Ticket Priority Analysis subagent with the Support Ticket Intelligence Auto-Launched Flow. This action enables the Agentforce subagent to invoke the Salesforce Flow and use its automated ticket processing capabilities.

Activities:

1. Accessed the Agentforce action configuration.
2. Created a new action based on the existing Salesforce Flow.
3. Selected the Support Ticket Intelligence Auto-Launched Flow.
4. Connected the Flow action to the Support Ticket Priority Analysis subagent.
5. Configured the action for use during Agentforce conversations.
6. Verified the action configuration.



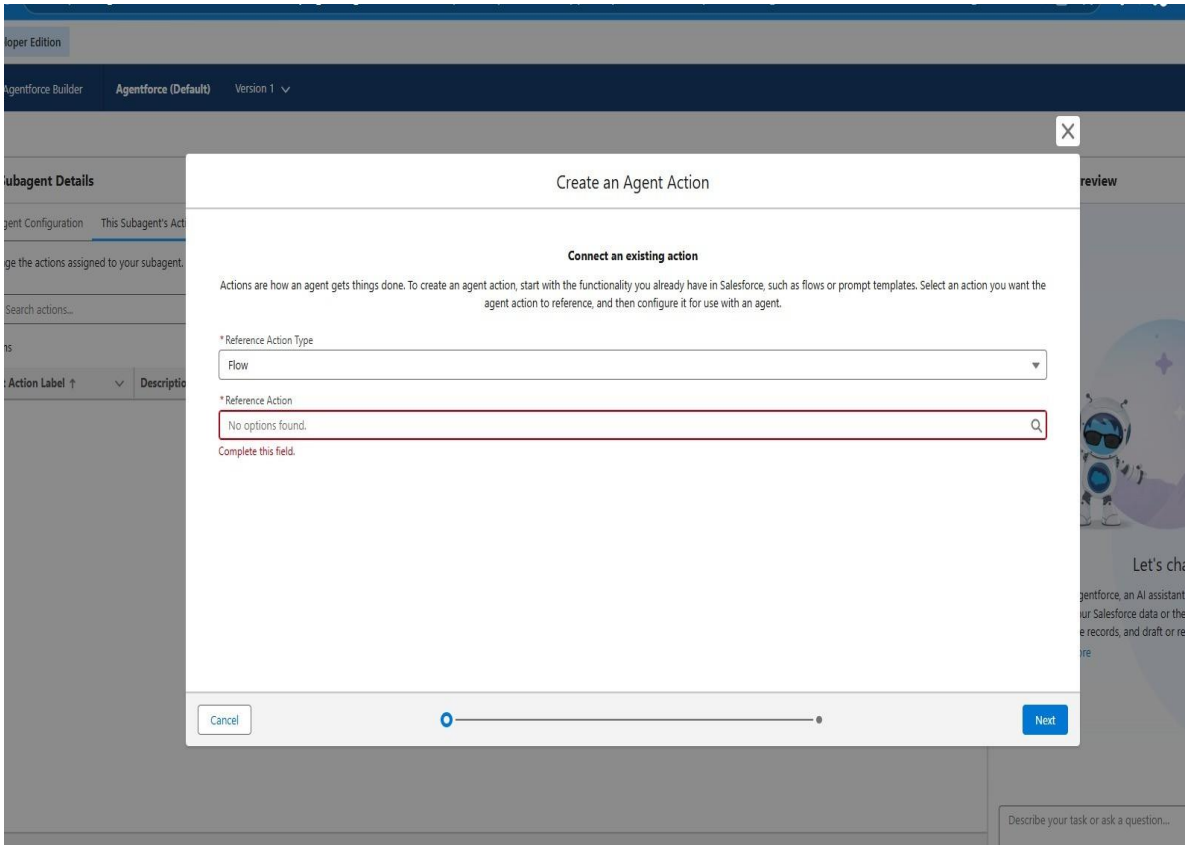
Step 4: Input and Output Variables

The input and output variables were configured for the Flow-based Agentforce action to enable data exchange between the Support Ticket Priority Analysis subagent and the Support Ticket Intelligence Flow. Input variables provide the required ticket information to the Flow, while output variables return the processed result to Agentforce.

Activities:

1. Configured the required input variables for the Agentforce action.

2. Mapped the input values to the corresponding Flow variables.
3. Configured the required output variables.
4. Mapped the Flow output to the Agentforce action response.
5. Verified the input and output mappings.



Step 5: Agentforce Conversation Response

The Agentforce conversation response was configured to present the result generated by the Support Ticket Priority Analysis subagent. After processing the ticket through the Flow-based action, Agentforce returns the relevant ticket priority and automated action information to the user in a conversational format.

Activities:

1. Invoked the Flow-based action through the Agentforce subagent.
2. Processed the support ticket information.
3. Received the output generated by the Flow.
4. Presented the ticket priority and relevant action in the Agentforce conversation.
5. Verified the final conversational response.

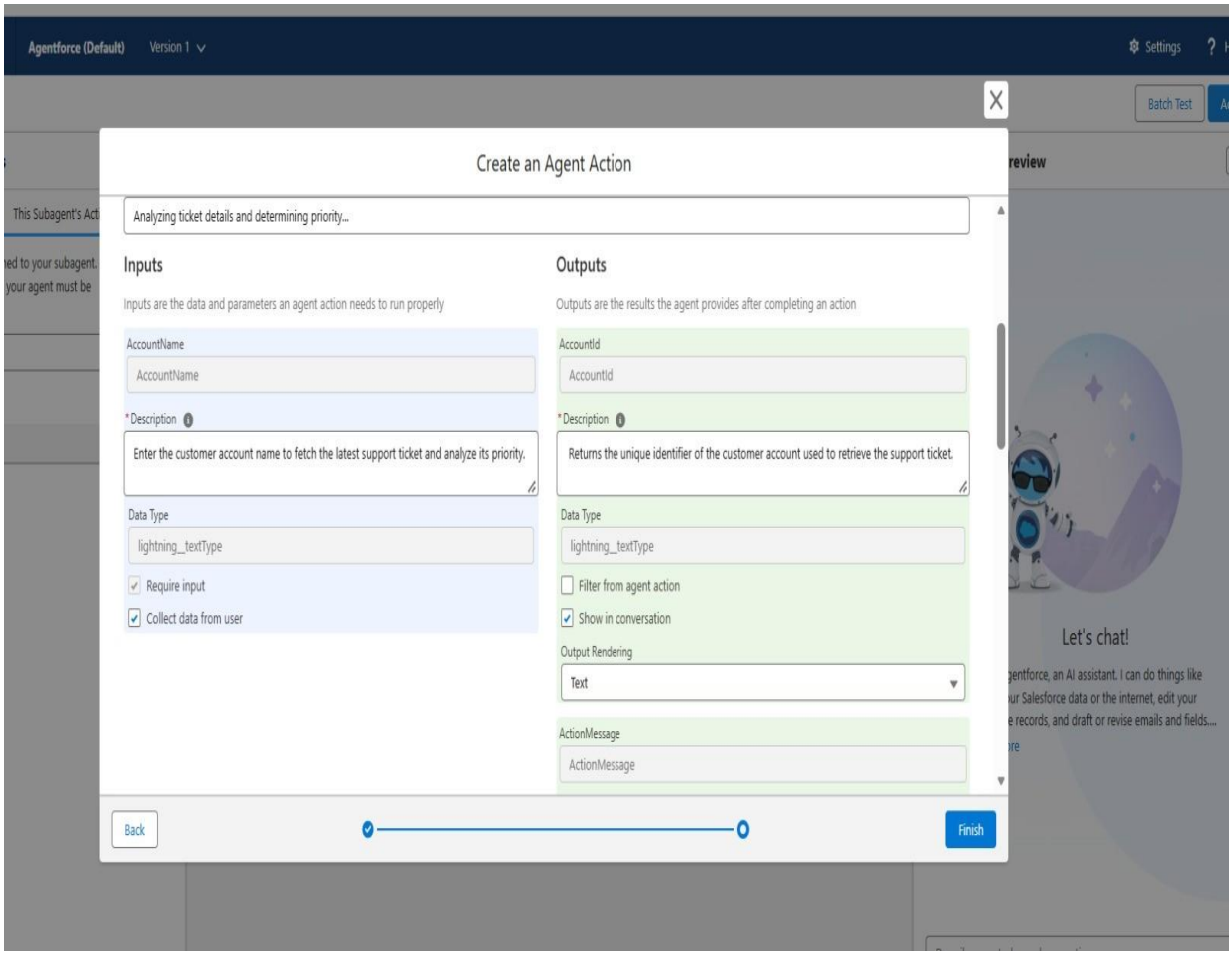
6.5 Phase 5 – Integration, Testing & Final Implementation

Step 1: Agentforce and Flow Integration

The Support Ticket Priority Analysis Agentforce subagent was integrated with the Support Ticket Intelligence Auto-Launched Flow through the configured Flow-based Agentforce action. This integration enables Agentforce to invoke the Flow, process support ticket information, classify the ticket priority, and perform the required automated actions.

Activities:

1. Connected the Agentforce subagent with the Support Ticket Intelligence Flow.
2. Configured the Flow-based Agentforce action.
3. Verified the input and output data mapping.
4. Tested the invocation of the Flow from Agentforce.
5. Confirmed that the Flow returned the expected result to Agentforce.

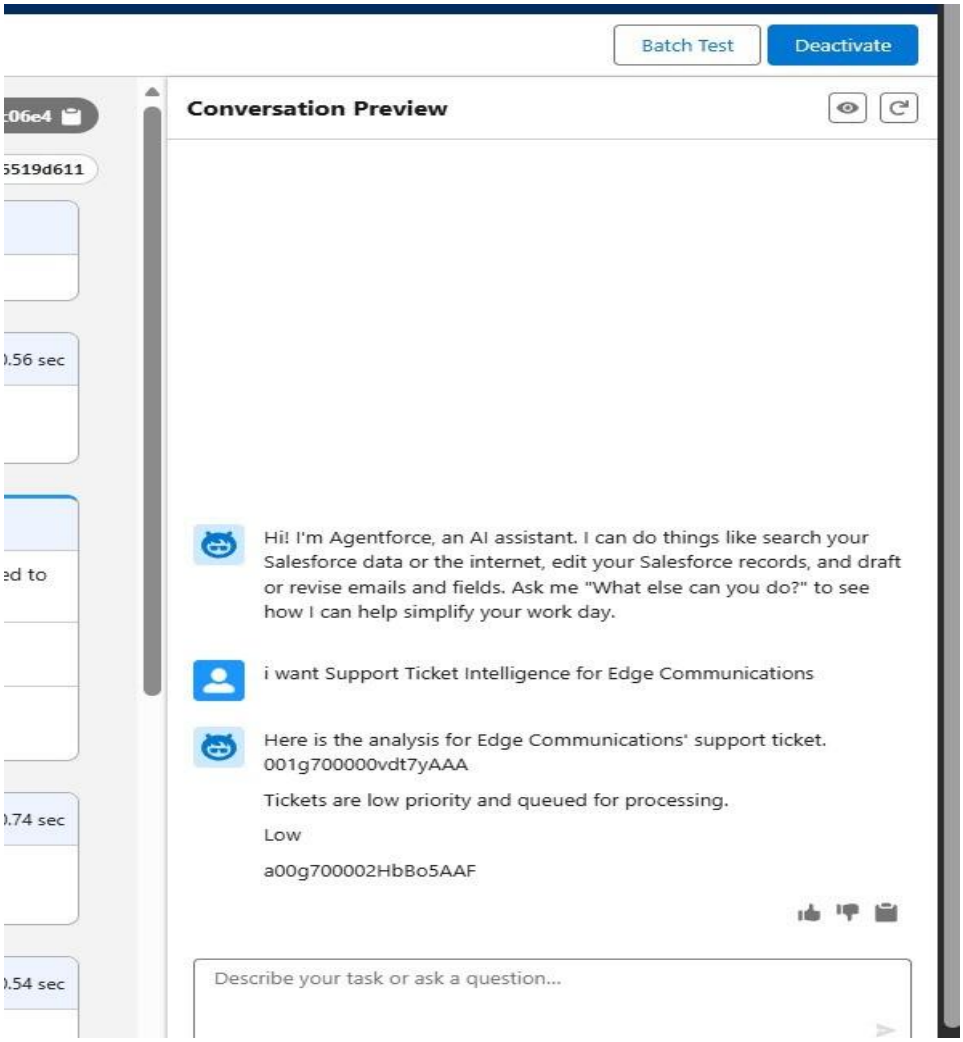


Step 2: Ticket Priority Validation

The ticket priority classification generated by the automation was validated using sample support ticket data. The ticket description was processed through the configured Flow and Agentforce components, and the resulting priority value was checked against the expected classification.

Activities:

1. Provided a sample support ticket for testing.
2. Processed the ticket through the Agentforce and Flow integration.
3. Verified the priority classification generated by the system.
4. Compared the generated priority with the expected result.
5. Confirmed that the priority value was correctly returned.



Step 3: Task Creation Validation

The automated Task creation functionality was validated by processing a high-priority support ticket through the configured Flow. The system was checked to ensure that a Task was automatically created when the ticket met the configured high-priority condition.

Activities:

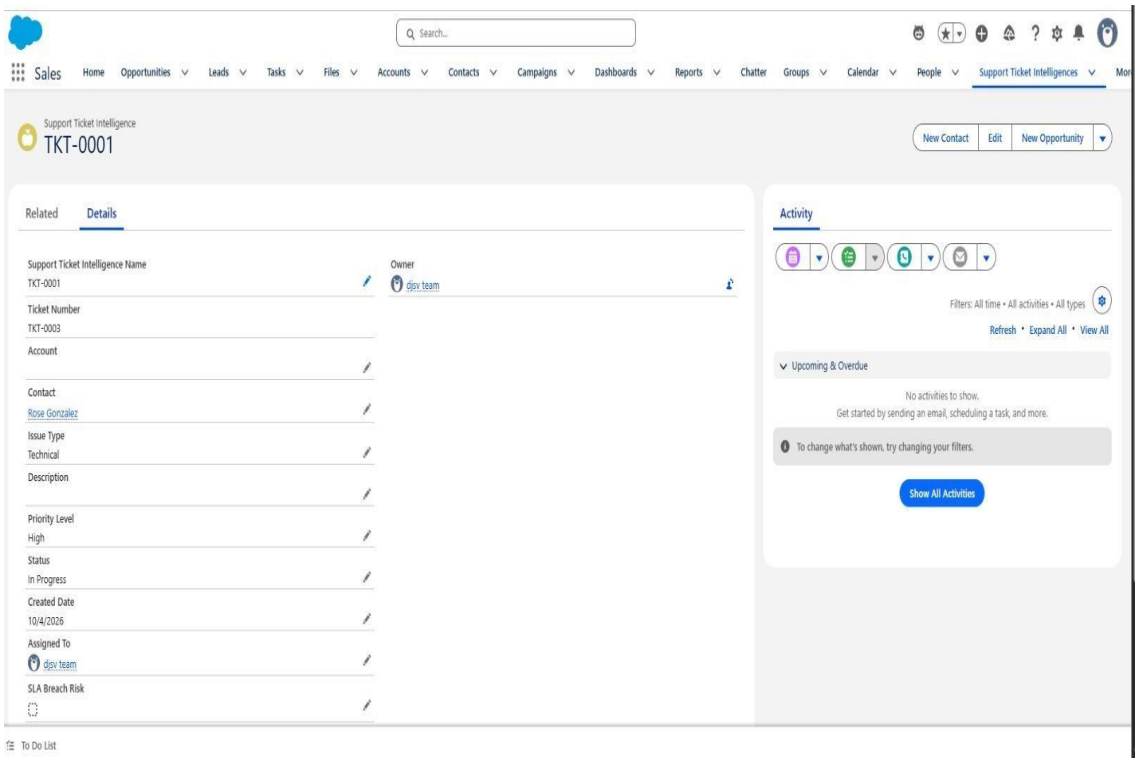
1. Submitted a sample high-priority support ticket.
2. Executed the configured Agentforce and Flow automation.
3. Verified that the high-priority condition was identified.
4. Checked whether a Task was automatically created.
5. Verified the Task details and associated ticket information.

Step 4: Agent Assignment Validation

The automated agent assignment functionality was validated to ensure that the support ticket and its corresponding Task were assigned to the configured Salesforce support user. The assignment result was verified after executing the Agentforce and Flow automation.

Activities:

1. Executed the ticket processing Flow through Agentforce.
2. Verified the support ticket assignment.
3. Checked the assigned Salesforce User.
4. Verified that the generated Task was assigned to the appropriate user.
5. Confirmed that the assignment information was correctly stored in Salesforce.

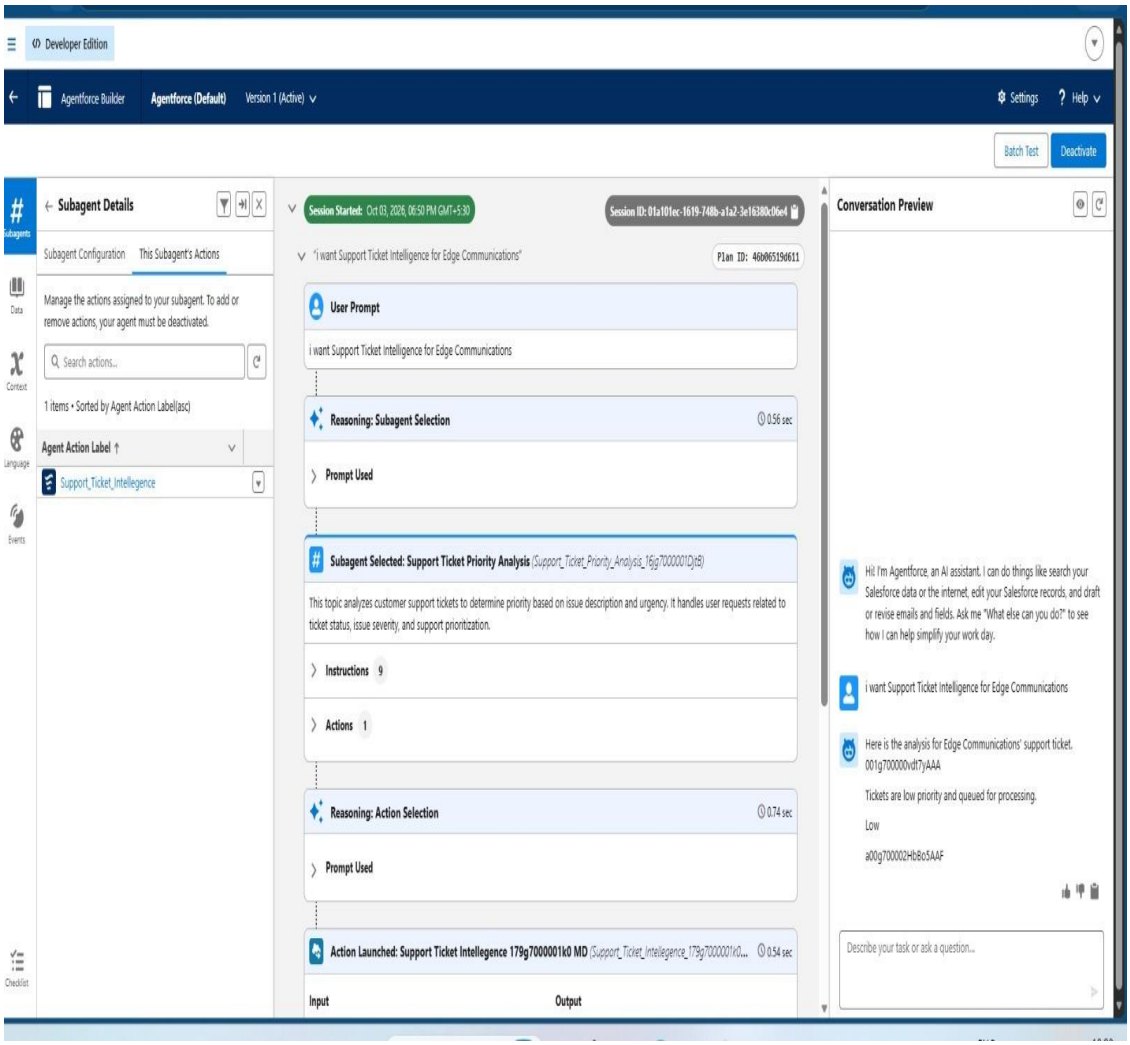


Step 5: Final Output Verification

The complete Support Ticket Intelligence automation system was verified from ticket submission to final Agentforce response. The integration between the custom Salesforce object, Flow, Task creation, agent assignment, and Agentforce response was checked to ensure that the complete workflow operated as intended.

Activities:

1. Submitted a sample support ticket through the configured system.
2. Verified retrieval and processing of the ticket information.
3. Confirmed the generated ticket priority.
4. Verified automatic Task creation for the applicable priority.
5. Confirmed assignment to the configured support agent.
6. Verified the final response generated through Agentforce.
7. Confirmed the overall automation workflow.



7. FUNCTIONAL & PERFORMANCE TESTING

7.1 Functional Testing

Functional testing was performed to verify that the major components of the **Customer Support Ticket Priority Prediction and Automated Assignment System** operate according to the defined requirements. The testing focused on ticket retrieval, priority classification, automated Task creation, agent assignment, and Agentforce response generation.

Functional Test Activities:

1. Verified support ticket retrieval from the Salesforce custom object.
2. Tested ticket description analysis and priority classification.

3. Verified High, Medium, and Low -priority paths.
4. Tested automatic Task creation for high-priority tickets.
5. Verified assignment of tickets to the configured support agent.
6. Tested the Agentforce conversation response.
7. Verified the final output returned by the integrated Flow and Agentforce components.

7.2 Functional Test Cases

The functional test cases were designed to verify the major operations of the Customer Support Ticket Priority Prediction and Automated Assignment System. Each test case was executed using sample Salesforce ticket data and the resulting behavior was compared with the expected outcome.

Test Case ID	Test Scenario	Input	Expected Result	Actual Result	Status
TC01	Account Retrieval	Valid Account Name	Correct Account record is retrieved	Account retrieved successfully	Pass
TC02	Latest Ticket Retrieval	Valid Account with support ticket	Latest associated ticket is retrieved	Ticket retrieved successfully	Pass
TC03	Ticket Description Analysis	Ticket description	Description is processed for priority analysis	Description analyzed successfully	Pass
TC04	High -Priority Classification	Urgent ticket description	Ticket is classified as High Priority	High Priority identified	Pass
TC05	Medium Priority Classification	Moderate issue description	Ticket is classified as Medium Priority	Medium Priority identified	Pass
TC06	Low- Priority Classification	Normal/low-severity issue description	Ticket is classified as Low Priority	Low Priority identified	Pass
TC07	Task Creation	High-priority ticket	Task is automatically created	Task created successfully	Pass
TC08	Agent Assignment	Processed support ticket	Ticket/Task is assigned to configured user	Agent assigned successfully	Pass
TC09	Agentforce Action	User requests ticket analysis	Flow-based Agentforce action is triggered	Action executed successfully	Pass

TC10	Final Response	Completed ticket analysis	Agentforce returns priority and action result	Response displayed successfully	Pass

7.3 Performance Testing

Performance testing was conducted to evaluate the responsiveness and execution behavior of the **Customer Support Ticket Priority Prediction and Automated Assignment System**. The testing focused on Flow execution, Agentforce response time, ticket processing, and automated task creation.

Performance Test Activities:

1. Executed the support ticket automation with sample ticket records.
2. Observed the Flow execution and processing behavior.
3. Verified the response generated by Agentforce after invoking the Flow-based action.
4. Checked the time required to retrieve and process ticket information.
5. Verified that automated Task creation and agent assignment were completed during the workflow.
6. Confirmed that the system produced the expected output without execution failure.

Performance Test Cases

Test Case ID	Performance Scenario	Expected Result	Observed Result	Status
PT01	Account and ticket retrieval	Records should be retrieved successfully	Records retrieved successfully	Pass
PT02	Priority classification	Priority should be determined without execution failure	Priority determined successfully	Pass
PT03	Agentforce action execution	Action should execute and return a response	Action executed successfully	Pass
PT04	High-priority Task creation	Task should be created during automation	Task created successfully	Pass

PT05	Agent assignment	User assignment should complete successfully	Agent assigned successfully	Pass
PT06	Final response generation	Final response should be returned correctly	Response generated successfully	Pass

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

The **Customer Support Ticket Priority Prediction and Automated Assignment System** provides several benefits by combining Salesforce Flow automation with Agentforce.

1. **Automated Ticket Prioritization**
   Support tickets can be classified into High, Medium, and Low priority based on the ticket description.
2. **Reduced Manual Work**
   The automated Flow reduces the need for manual ticket processing and follow-up activities.
3. **Automatic Task Creation**
   A Task can be automatically created for high-priority support tickets.
4. **Automated Agent Assignment**
   The system assigns the ticket or related Task to the configured support user.
5. **Faster Ticket Processing**
   Automation helps process support requests through a predefined workflow.
6. **Agentforce Integration**
   Users can interact with the system through a conversational Agentforce interface.
7. **Centralized Data Management**
   Customer, Account, Contact, ticket, priority, and assignment information can be managed within Salesforce.
8. **Consistent Processing**
   The configured Flow provides a standardized process for handling support tickets.

8.2 Disadvantages

1. **Salesforce Dependency**
   The system depends on the Salesforce platform and its available features.

2. **Configuration Complexity**
   Setting up custom objects, Flow automation, and Agentforce components requires configuration knowledge.
3. **Data Quality Dependency**
   The accuracy of ticket processing depends on the quality and completeness of the ticket information.
4. **Limited Classification Rules**
   The current implementation uses predefined priority classification logic and may require additional rules for more complex support scenarios.
5. **Maintenance Requirement**
   Flow, Agentforce actions, and configurations may require updates when business requirements change.
6. **Testing Requirements**
   Different types of customer issues need to be tested to ensure consistent priority classification and automation behavior.

9. CONCLUSION

The **Customer Support Ticket Priority Prediction and Automated Assignment System Using Agentforce** was successfully designed and implemented using Salesforce and Agentforce technologies. The system integrates a custom **Support Ticket Intelligence** object, Salesforce Flow automation, and an Agentforce subagent to process customer support tickets.

The implemented solution analyzes ticket information, determines the appropriate priority level, creates Tasks for high-priority tickets, and assigns the required support user. The integration of Agentforce with the Salesforce Flow enables users to interact with the system through a conversational interface while the underlying automation performs the required processing.

Functional and performance testing was carried out using sample support ticket records. The test results verified ticket retrieval, priority classification, automated Task creation, agent assignment, and final Agentforce response generation.

Overall, the project demonstrates how Salesforce automation and Agentforce can be integrated to create a structured and intelligent approach for customer support ticket management. The implemented system provides a foundation that can be further enhanced with advanced classification, analytics, and intelligent customer support capabilities.

10. FUTURE SCOPE

The **Customer Support Ticket Priority Prediction and Automated Assignment System Using Agentforce** can be further enhanced with additional intelligent features and advanced automation capabilities.

1. **Machine Learning-Based Priority Prediction**
   A machine learning model can be integrated to predict ticket priority using historical support ticket data and customer issue patterns.

2. **Advanced Natural Language Processing**
   NLP techniques can be used to understand customer descriptions, identify important keywords, and improve ticket classification.

3. **Automatic Agent Selection**
   The system can be enhanced to select support agents based on their skills, workload, availability, and ticket category.

4. **Sentiment Analysis**
   Customer sentiment can be analyzed to identify frustrated or dissatisfied customers and provide appropriate priority handling.

5. **Real-Time Notifications**
   Automated notifications can be sent to support agents when high-priority or critical tickets are created.

6. **Advanced Dashboard and Analytics**
   Salesforce dashboards can be developed to monitor ticket volume, priority distribution, agent workload, resolution time, and system performance.

7. **Multi-Channel Customer Support**
   The system can be extended to process tickets received through email, chat, web forms, and other customer communication channels.

8. **Continuous Learning and Improvement**
   Historical ticket outcomes can be analyzed to improve priority classification rules and make the system more adaptive to changing customer support requirements.

9. **Escalation Automation**
   An automated escalation mechanism can be introduced for unresolved high-priority tickets that exceed predefined response or resolution times.

10. **Integration with External Systems**
    The solution can be integrated with external CRM, communication, analytics, and enterprise applications to create a more comprehensive customer support ecosystem.

11. APPENDIX

The appendix provides additional information related to the configuration, implementation, and testing of the **Customer Support Ticket Priority Prediction and Automated Assignment System Using Agentforce**.

A. Custom Object – Support Ticket Intelligence

The **Support Ticket Intelligence** custom object was created to store and manage customer support ticket information. The object contains the required fields and relationships used for ticket processing, priority classification, and automation.

B. Salesforce Records – Account, Contact, User and Task

Salesforce **Account, Contact, User, and Task** records were configured to support customer identification, ticket association, agent assignment, and automated follow-up activities.

C. Auto-Launched Flow – Support Ticket Intelligence

The **Support Ticket Intelligence Auto-Launched Flow** was developed to automate ticket processing. The Flow performs account retrieval, latest ticket retrieval, ticket description analysis, priority classification, Task creation, agent assignment, and final response generation.

D. Agentforce Subagent – Support Ticket Priority Analysis

The **Support Ticket Priority Analysis** Agentforce subagent was configured to analyze customer support ticket information and determine the appropriate priority based on the defined instructions.

E. Agentforce Action – Flow-Based Support Ticket Intelligence

A Flow-based Agentforce action was configured to connect the Agentforce subagent with the **Support Ticket Intelligence** Flow. Input and output variables were mapped to enable communication between the Agentforce component and Salesforce Flow.

F. Implementation

The complete system was implemented by integrating the Salesforce custom object, Flow automation, Agentforce subagent, and Flow-based Agentforce action. The integrated solution processes support tickets, determines priority, creates required Tasks, and assigns support users.

G. Testing

The implemented system was tested using sample support ticket records. Functional testing verified ticket retrieval, priority classification, Task creation, agent assignment,

action execution, and final response generation. Performance testing was also carried out to verify the successful execution of the overall workflow.

THANK YOU